Sprint Report

Portfolio Task 2/3/4

Unit code: COS40005/EAT40005

Unit name: Computing/Engineering Technology Project A

Submission date: 28th of November, 2025

Group 7:

Daniel Tiong (102777801)

Alif Harriz (102782711)

Ashley Jong (102780087)

Basil Agas (102778888)

Edison Ho (102779496)

1. Contribution Summary (This Sprint)

All team members should complete the following table together. You can provide a yes/no response or a brief comment where appropriate.

Team Member	Daniel Tiong	Alif Harriz	Ashley Jong	Basil Agas	Edison Ho
Contribution Finished tasks in time or on time with an acceptable quality	Yes	Yes	Yes	Yes	Yes
Initiatives Self-assigned tasks and proactively found solutions to problems	Yes	Yes	Yes	Yes	Yes
Communication Attended all team meetings	Yes	Yes	Yes	Yes	Yes
Communication Attended all supervisor meetings	Yes	Yes	Yes	Yes	Yes
Communication Attended all client meetings	Yes	Yes	Yes	Yes	Yes
Communication Replied every time within an acceptable timeframe	Yes	Yes	Yes	Yes	Yes
Communication	Yes	Yes	Yes	Yes	Yes

Kept the team updated about the status					
Respect **The student respected others and listened to different opinions**	Yes	Yes	Yes	Yes	Yes

2. Contribution Details (This Sprint)

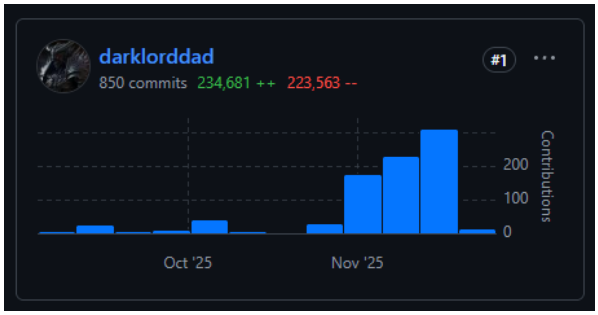
In this section, every member of the team should write a summary of what they have contributed during this sprint. All contributions must be supported by evidence, such as the number of GitHub commits, a screenshot of developed screens, a summary/outcome of research, etc.

2.1. Daniel (Project Lead and AI Engineer)

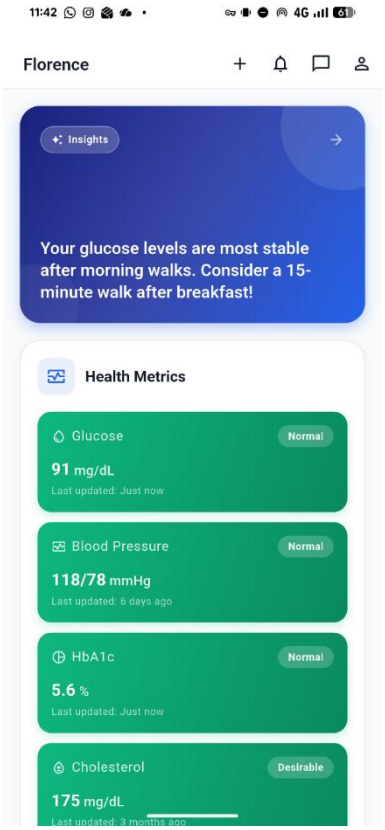
Planned: Work on LangChain Agents.

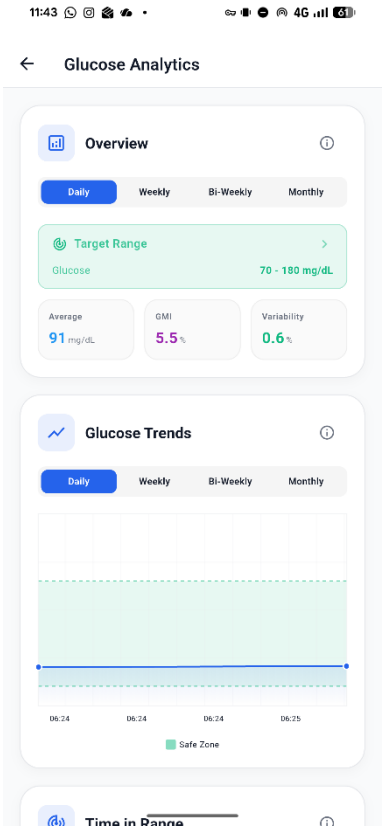
Actual: I pivoted to **Integration and Architecture**. I deleted legacy AI files and focused on making the patient dashboard fully functional with our backend service. This aligns perfectly with the “Strategic Pivot” narrative in the report.

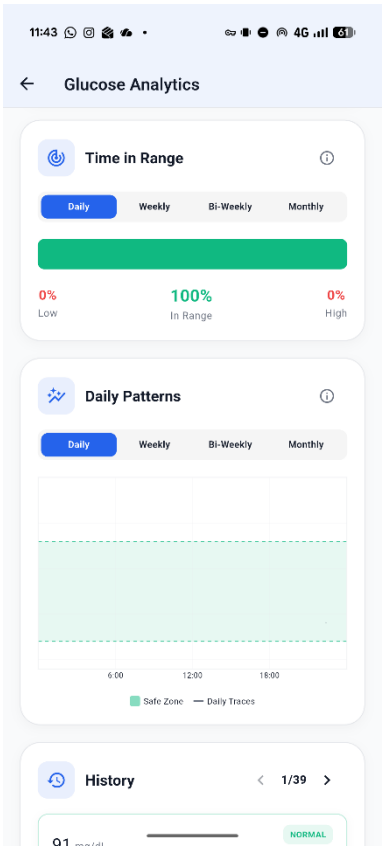
Evidence of Work

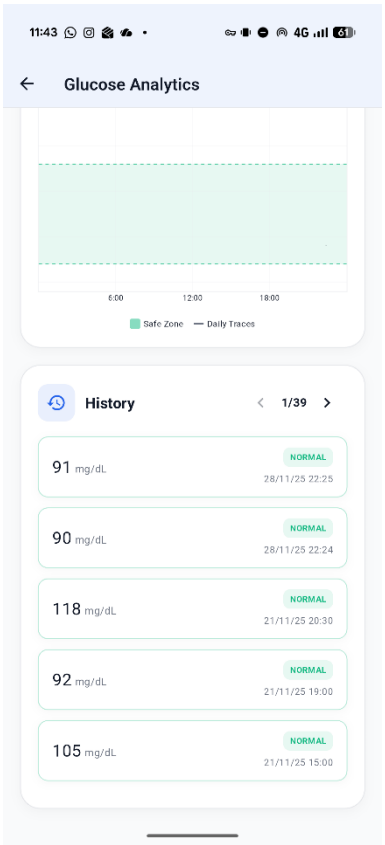


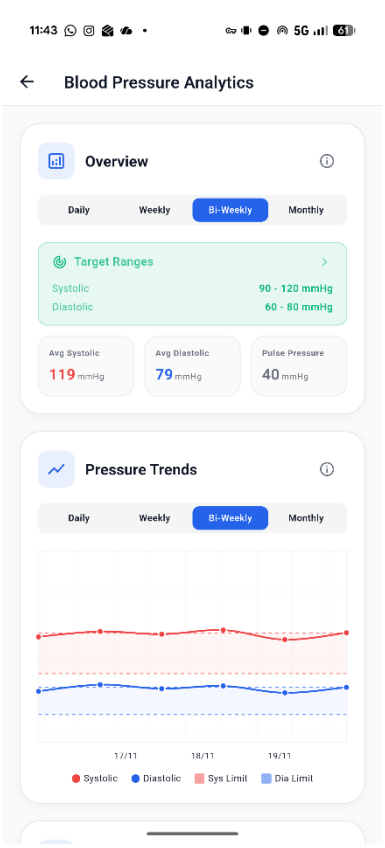
Screenshot of the Application

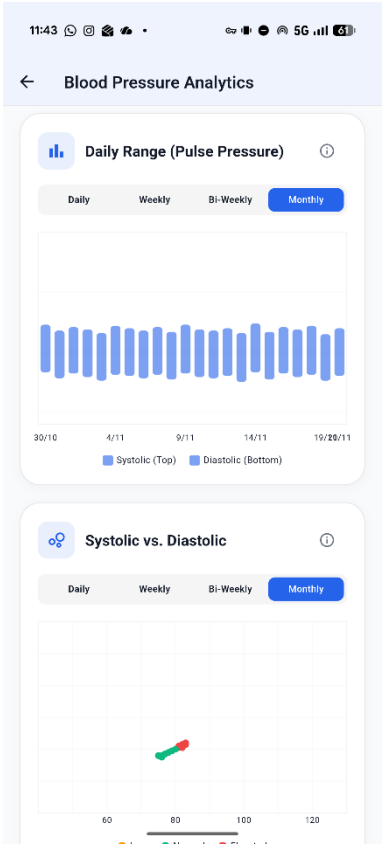


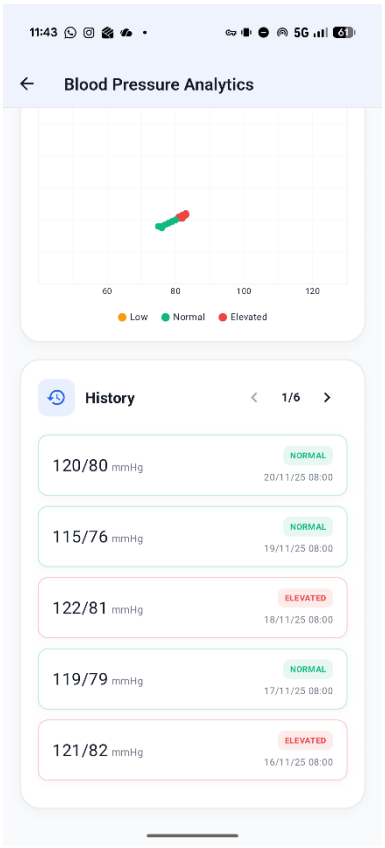


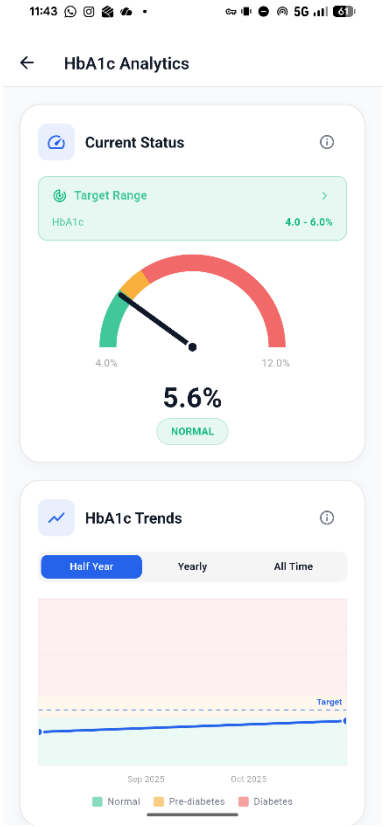


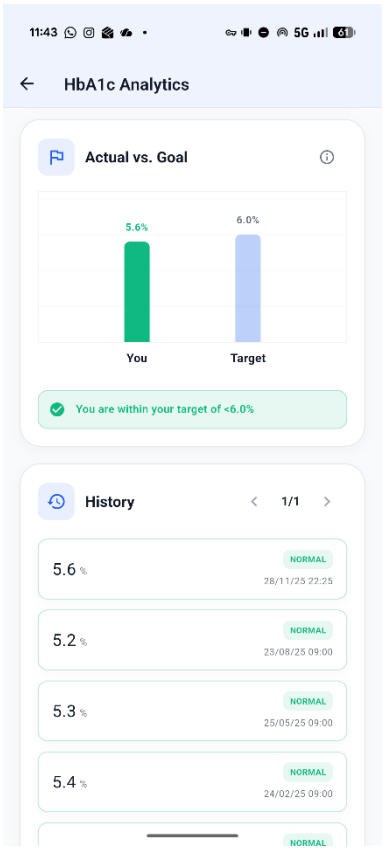


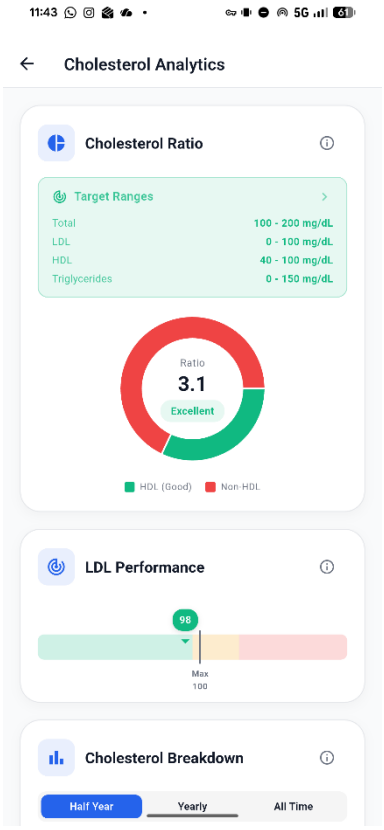

11:43
Blood Pressure Analytics
Overview
Daily
Weekly
Bi-Weekly
Monthly
Target Ranges
Systolic
90 - 120 mmHg
Diastolic
60 - 80 mmHg
Avg Systolic
119 mmHg
Avg Diastolic
79 mmHg
Pulse Pressure
40 mmHg
Pressure Trends
Daily
Weekly
Bi-Weekly
Monthly
17/11
18/11
19/11
Systolic
Diastolic
Sys Limit
Dia Limit

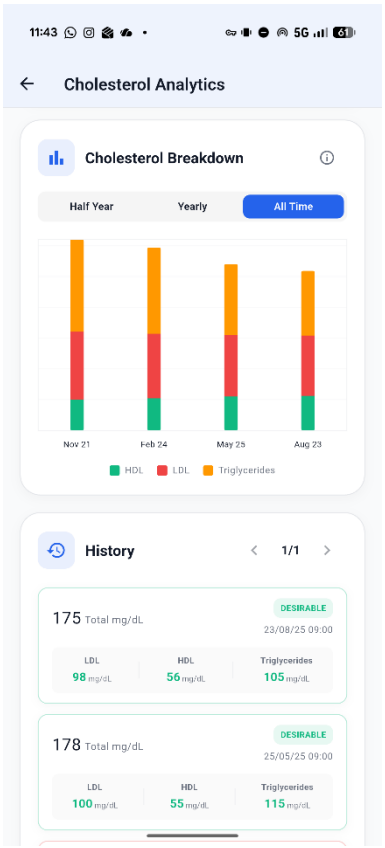

11:43
Blood Pressure Analytics
Daily Range (Pulse Pressure)
Daily
Weekly
Bi-Weekly
Monthly
30/10
4/11
9/11
14/11
19/20/11
Systolic (Top)
Diastolic (Bottom)
Systolic vs. Diastolic
Daily
Weekly
Bi-Weekly
Monthly
60
80
100
120

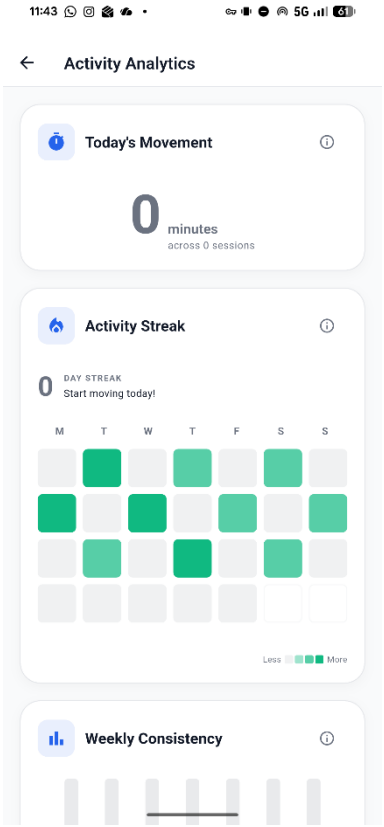

11:43
Blood Pressure Analytics
60
80
100
120
Low
Normal
Elevated
History
1/6
120/80 mmHg
NORMAL
20/11/25 08:00
115/76 mmHg
NORMAL
19/11/25 08:00
122/81 mmHg
ELEVATED
18/11/25 08:00
119/79 mmHg
NORMAL
17/11/25 08:00
121/82 mmHg
ELEVATED
16/11/25 08:00

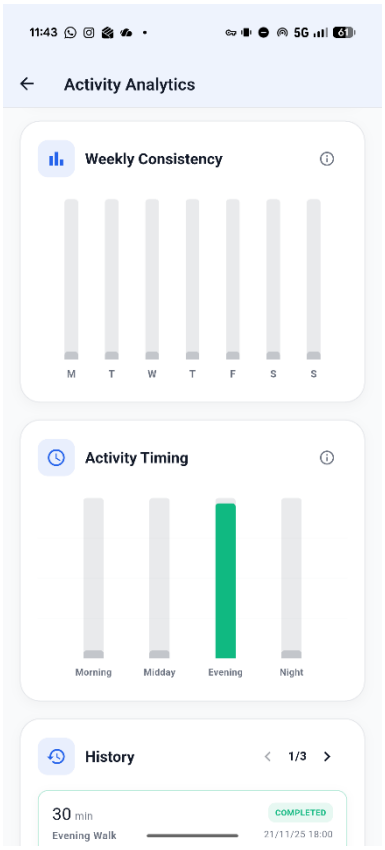

11:43
HbA1c Analytics
Current Status
Target Range
HbA1c
4.0 - 6.0%
4.0%
12.0%
5.6%
NORMAL
HbA1c Trends
Half Year
Yearly
All Time
Target
Sep 2025
Oct 2025
Normal
Pre-diabetes
Diabetes

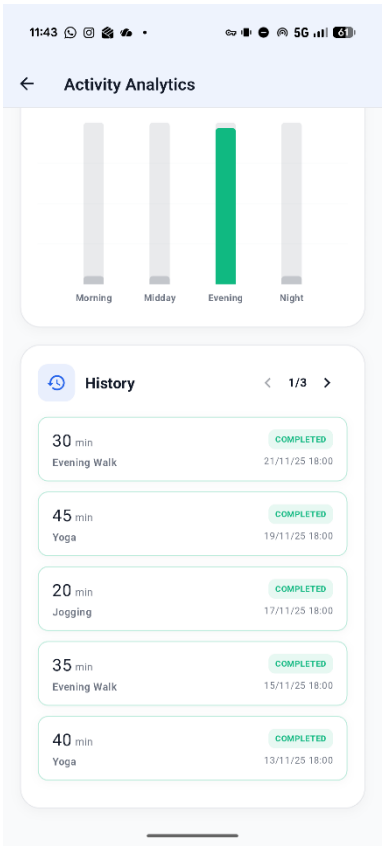

11:43
HbA1c Analytics
Actual vs. Goal
5.6%
6.0%
You
Target
You are within your target of <6.0%
History
1/1
5.6 %
NORMAL
28/11/25 22:25
5.2 %
NORMAL
23/08/25 09:00
5.3 %
NORMAL
25/05/25 09:00
5.4 %
NORMAL
24/02/25 09:00
NORMAL

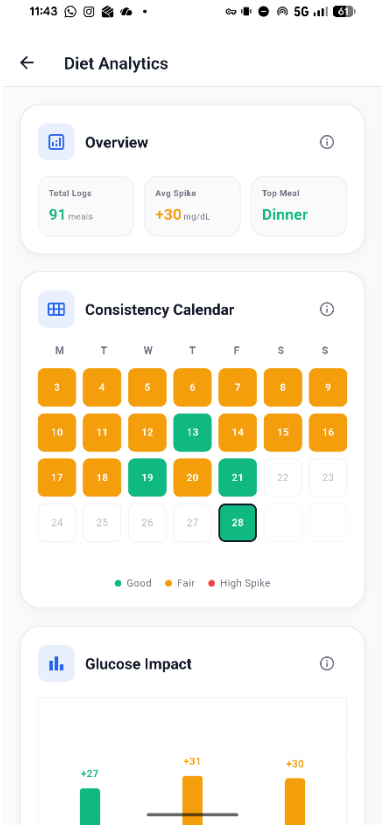

11:43
Cholesterol Analytics
Cholesterol Ratio
Target Ranges
Total 100 - 200 mg/dL
LDL 0 - 100 mg/dL
HDL 40 - 100 mg/dL
Triglycerides 0 - 150 mg/dL
Ratio
3.1
Excellent
HDL (Good)
Non-HDL
LDL Performance
98
Max
100
Cholesterol Breakdown
Half Year
Yearly
All Time

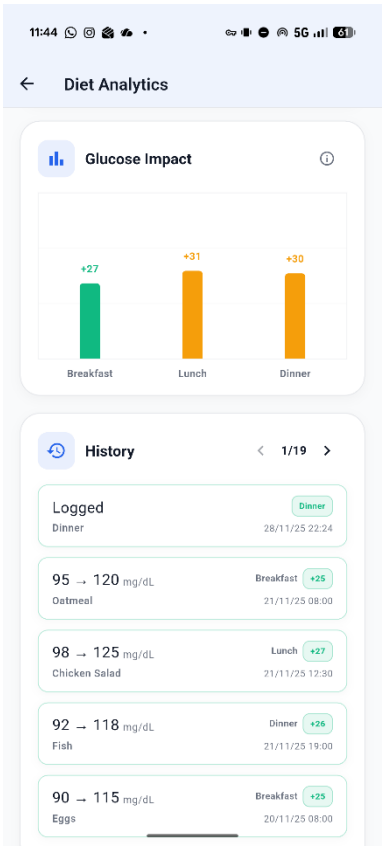

11:43
Cholesterol Analytics
Cholesterol Breakdown
Half Year
Yearly
All Time
Nov 21
Feb 24
May 25
Aug 23
HDL
LDL
Triglycerides
History
1/1
175 Total mg/dL
DESIRABLE
23/08/25 09:00
LDL 98 mg/dL
HDL 56 mg/dL
Triglycerides 105 mg/dL
178 Total mg/dL
DESIRABLE
25/05/25 09:00
LDL 100 mg/dL
HDL 55 mg/dL
Triglycerides 115 mg/dL

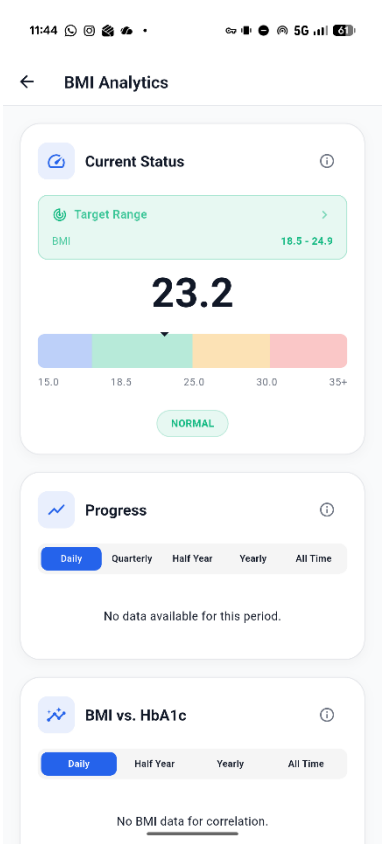

11:43
Activity Analytics
Today's Movement
0
minutes
across 0 sessions
Activity Streak
0
DAY STREAK
Start moving today!
M T W T F S S
Less
More
Weekly Consistency

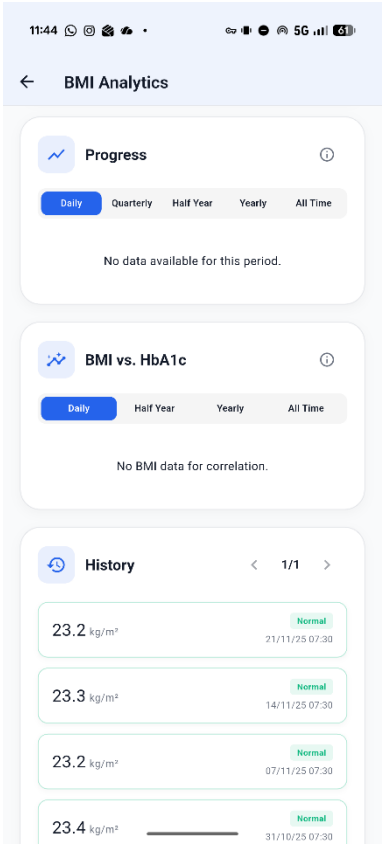

11:43
Activity Analytics
Weekly Consistency
M T W T F S S
Activity Timing
Morning Midday Evening Night
History
1/3
30 min
Evening Walk
COMPLETED
21/11/25 18:00

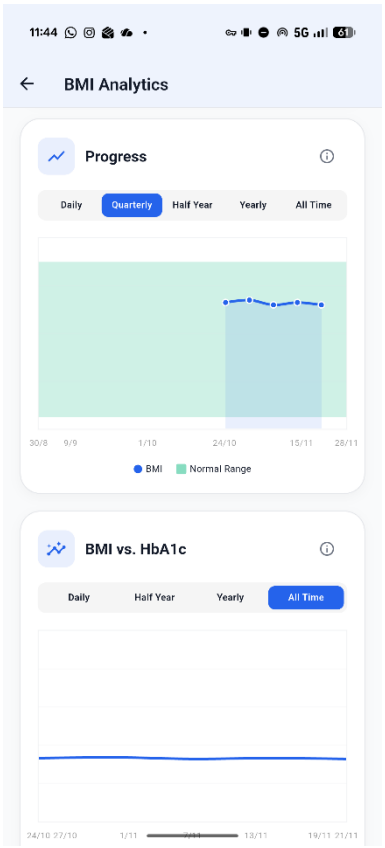

11:43
Activity Analytics
Morning Midday Evening Night
History
1/3
30 min
Evening Walk
COMPLETED
21/11/25 18:00
45 min
Yoga
COMPLETED
19/11/25 18:00
20 min
Jogging
COMPLETED
17/11/25 18:00
35 min
Evening Walk
COMPLETED
15/11/25 18:00
40 min
Yoga
COMPLETED
13/11/25 18:00

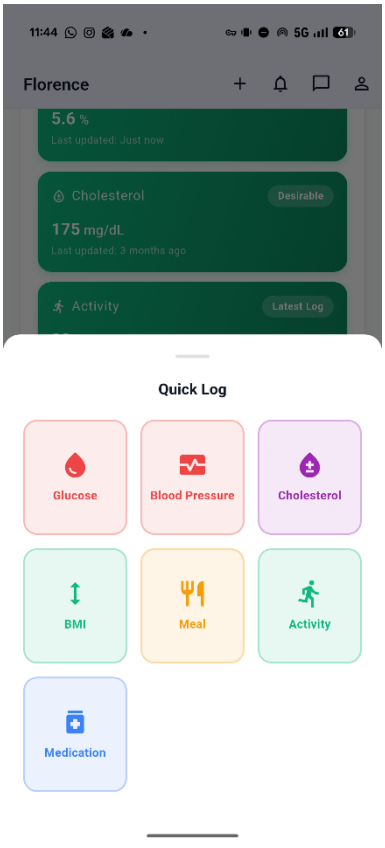

11:43
Diet Analytics
Overview
Total Logs
91 meals
Avg Spike
+30 mg/dL
Top Meal
Dinner
Consistency Calendar
M T W T F S S
3 4 5 6 7 8 9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28
Good Fair High Spike
Glucose Impact
+27 +31 +30

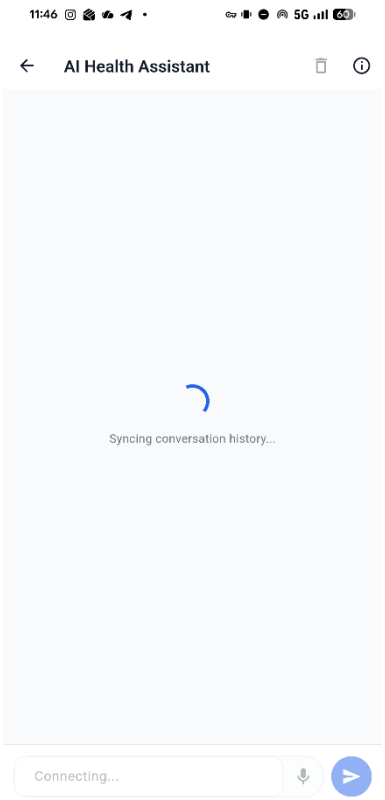

11:44
Diet Analytics
Glucose Impact
+27 +31 +30
Breakfast Lunch Dinner
History
1/19
Logged
Dinner
Dinner
28/11/25 22:24
95 → 120 mg/dL
Oatmeal
Breakfast +25
21/11/25 08:00
98 → 125 mg/dL
Chicken Salad
Lunch +27
21/11/25 12:30
92 → 118 mg/dL
Fish
Dinner +26
21/11/25 19:00
90 → 115 mg/dL
Eggs
Breakfast +25
20/11/25 08:00

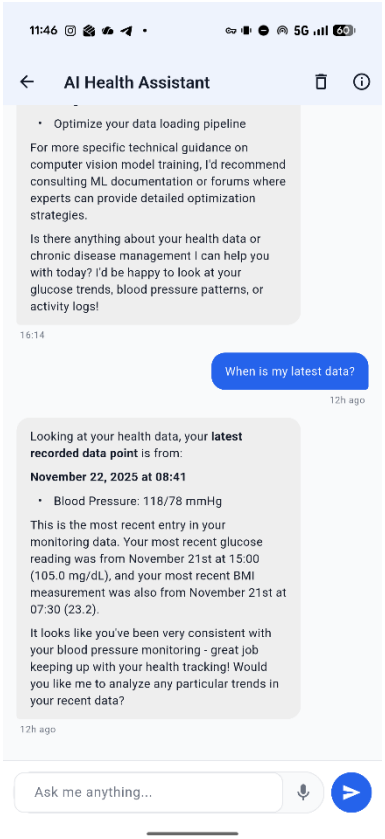

11:44
BMI Analytics
Current Status
Target Range
BMI
18.5 - 24.9
23.2
15.0
18.5
25.0
30.0
35+
NORMAL
Progress
Daily
Quarterly
Half Year
Yearly
All Time
No data available for this period.
BMI vs. HbA1c
Daily
Half Year
Yearly
All Time
No BMI data for correlation.

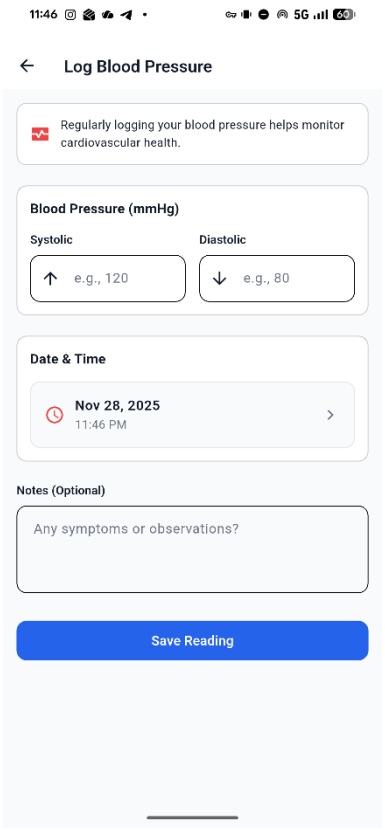

11:44
BMI Analytics
Progress
Daily
Quarterly
Half Year
Yearly
All Time
No data available for this period.
BMI vs. HbA1c
Daily
Half Year
Yearly
All Time
No BMI data for correlation.
History
1/1
23.2 kg/m²
Normal
21/11/25 07:30
23.3 kg/m²
Normal
14/11/25 07:30
23.2 kg/m²
Normal
07/11/25 07:30
23.4 kg/m²
Normal
31/10/25 07:30


11:44
BMI Analytics
Progress
Daily
Quarterly
Half Year
Yearly
All Time
30/8
9/9
1/10
24/10
15/11
28/11
BMI
Normal Range
BMI vs. HbA1c
Daily
Half Year
Yearly
All Time


11:44
Florence
5.6 %
Last updated: Just now
Cholesterol
Desirable
175 mg/dL
Last updated: 3 months ago
Activity
Latest Log
Quick Log
Glucose
Blood Pressure
Cholesterol
BMI
Meal
Activity
Medication


11:46
AI Health Assistant
Syncing conversation history...
Connecting...

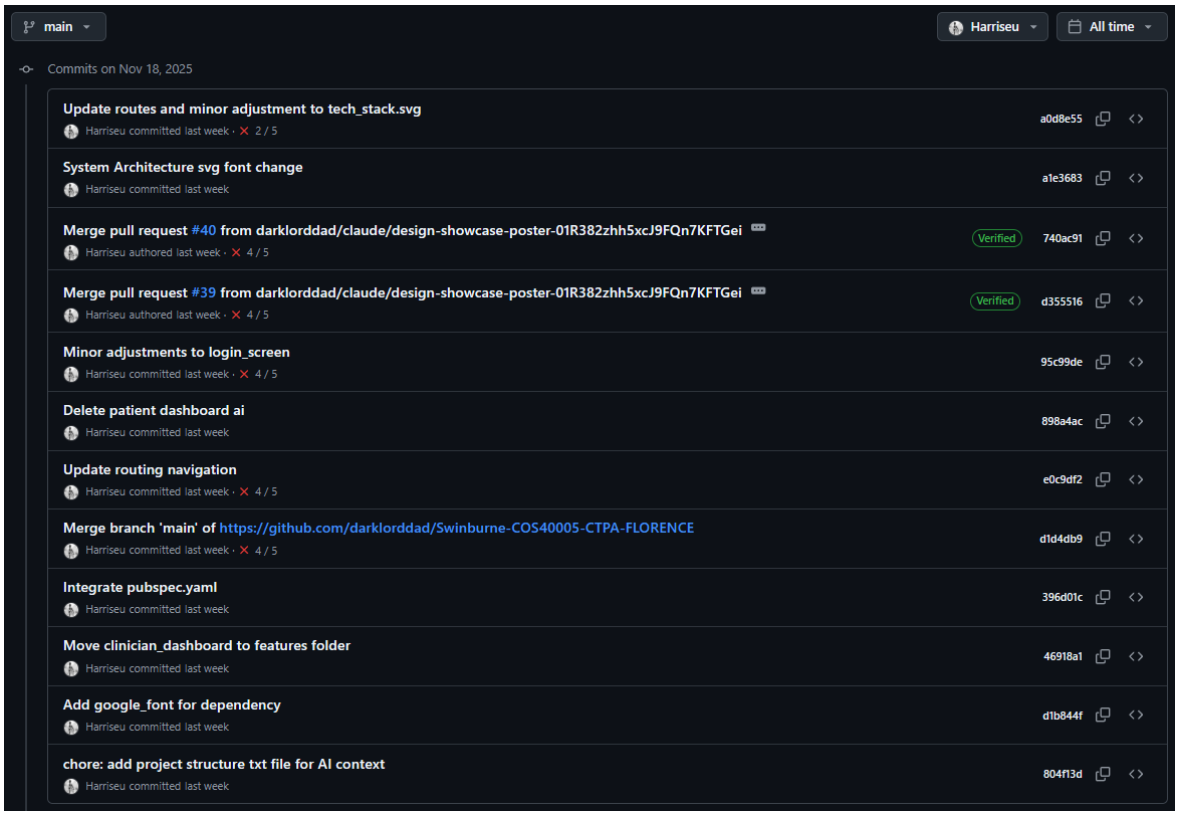

11:46
AI Health Assistant
• Optimize your data loading pipeline
For more specific technical guidance on computer vision model training, I'd recommend consulting ML documentation or forums where experts can provide detailed optimization strategies.
Is there anything about your health data or chronic disease management I can help you with today? I'd be happy to look at your glucose trends, blood pressure patterns, or activity logs!
16:14
When is my latest data?
12h ago
Looking at your health data, your latest recorded data point is from:
November 22, 2025 at 08:41
• Blood Pressure: 118/78 mmHg
This is the most recent entry in your monitoring data. Your most recent glucose reading was from November 21st at 15:00 (105.0 mg/dL), and your most recent BMI measurement was also from November 21st at 07:30 (23.2).
It looks like you've been very consistent with your blood pressure monitoring - great job keeping up with your health tracking! Would you like me to analyze any particular trends in your recent data?
12h ago
Ask me anything...

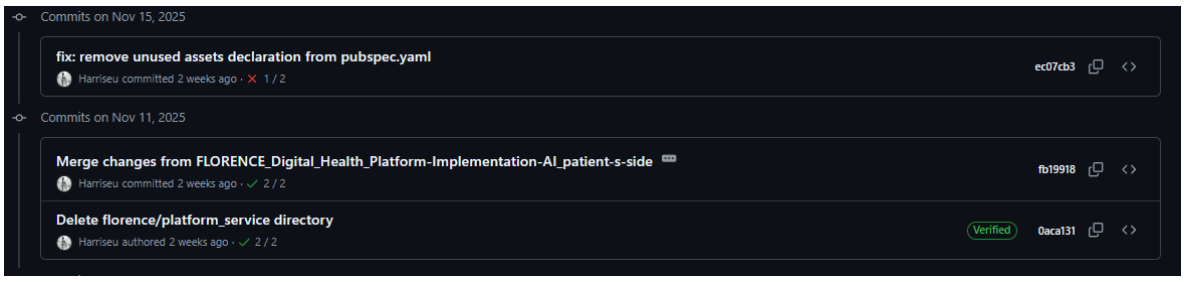

11:46
Log Blood Pressure
Regularly logging your blood pressure helps monitor cardiovascular health.
Blood Pressure (mmHg)
Systolic
e.g., 120
Diastolic
e.g., 80
Date & Time
Nov 28, 2025
11:46 PM
Notes (Optional)
Any symptoms or observations?
Save Reading

2.2. Alif (Frontend Developer - Patient Dashboard)

Evidence of Work

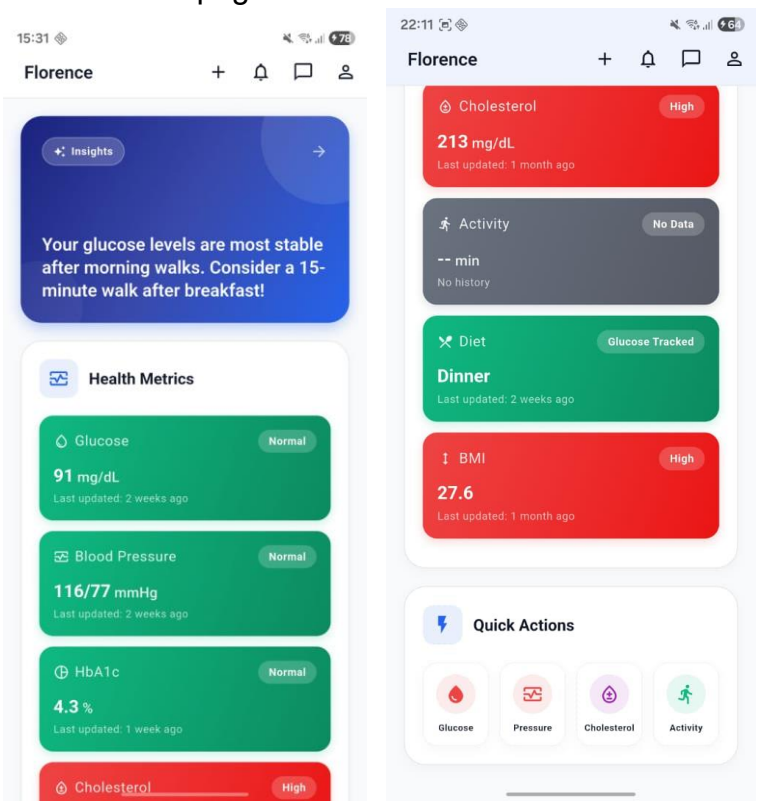
GitHub Commits



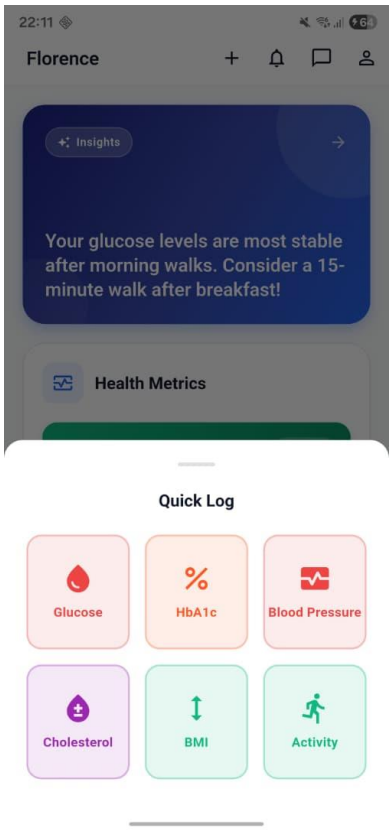


Screenshot of Application

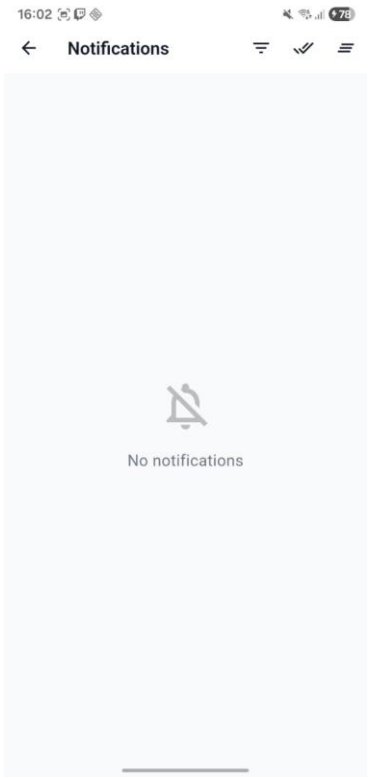
Patient Homepage



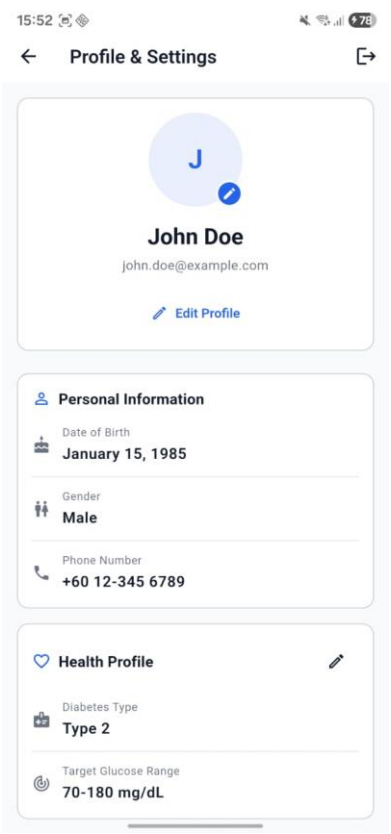
Patient Quick Log

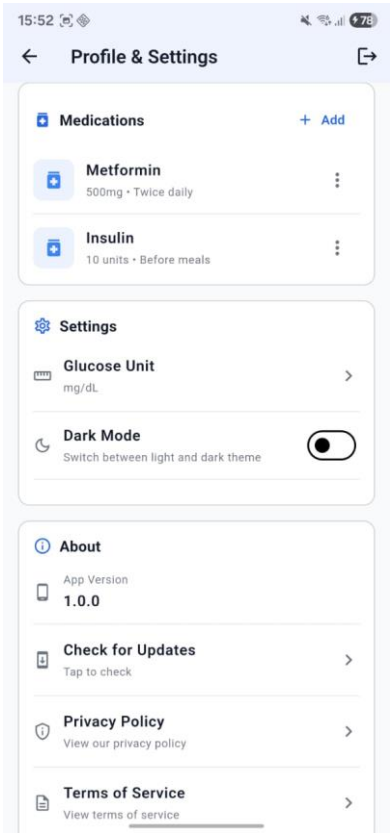


Patient Notifications

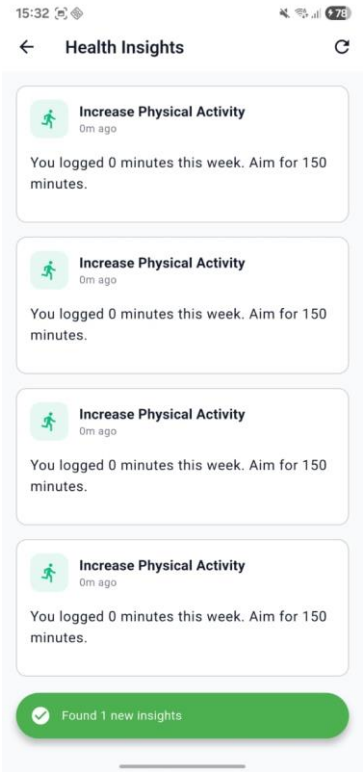


Patient Profile

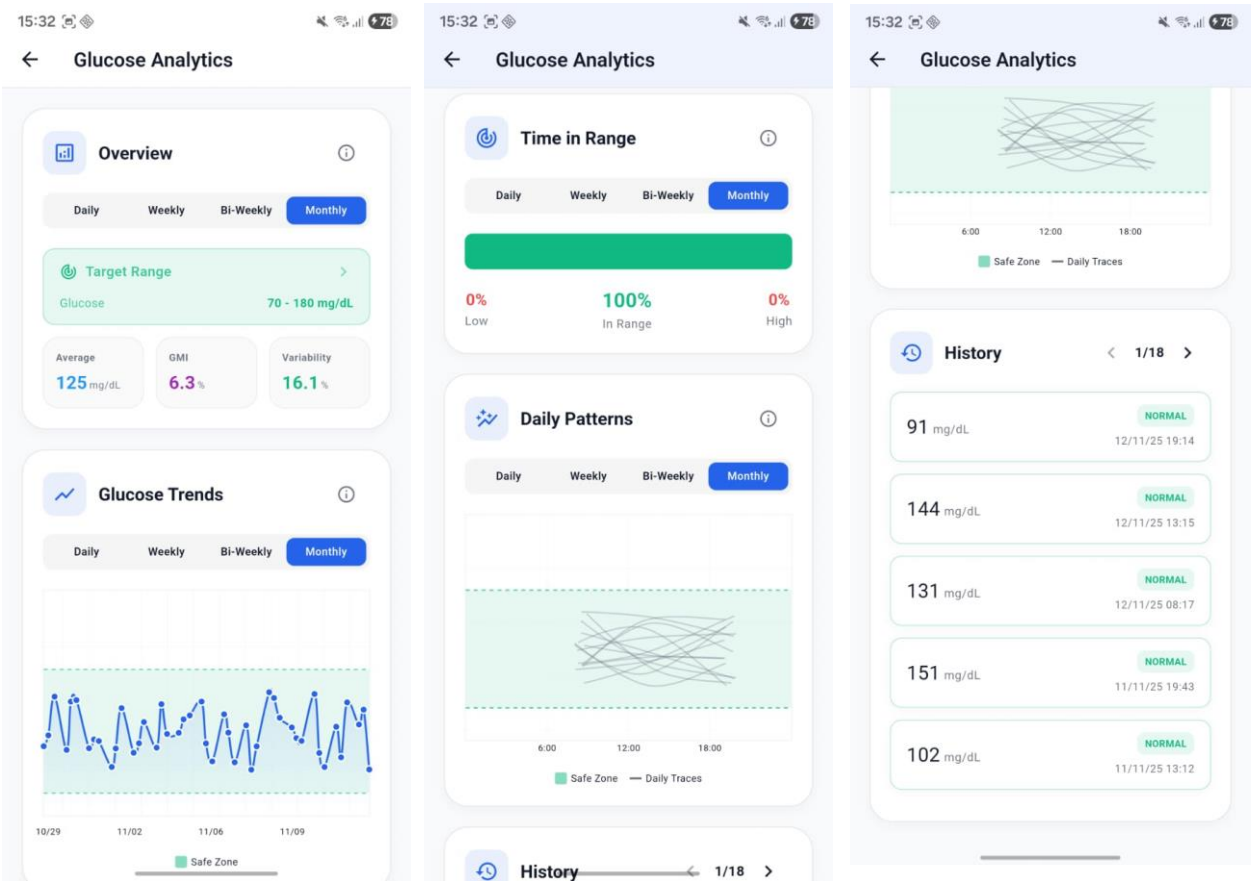




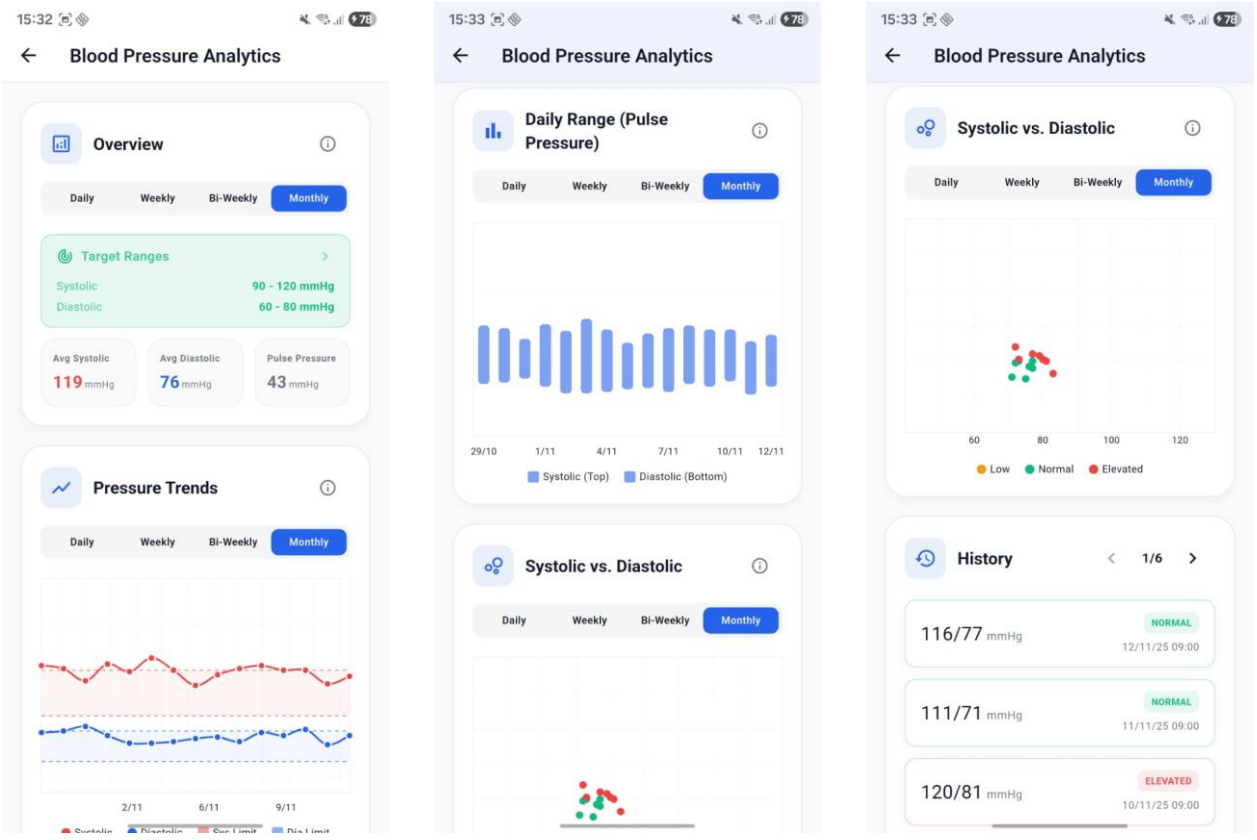
Patient Insights



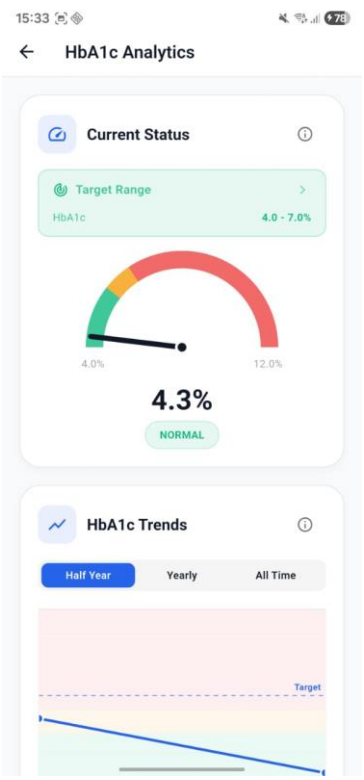
Patient Glucose Analytics

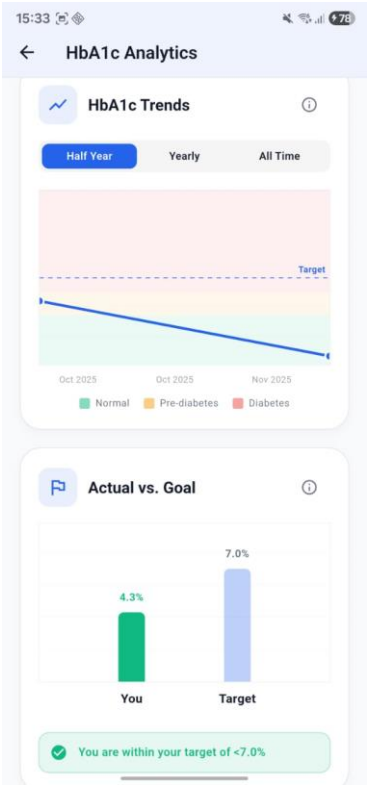


Patient Blood Pressure Analytics

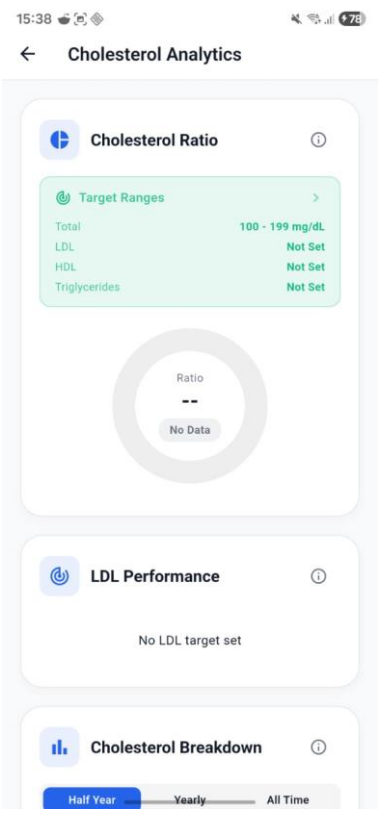


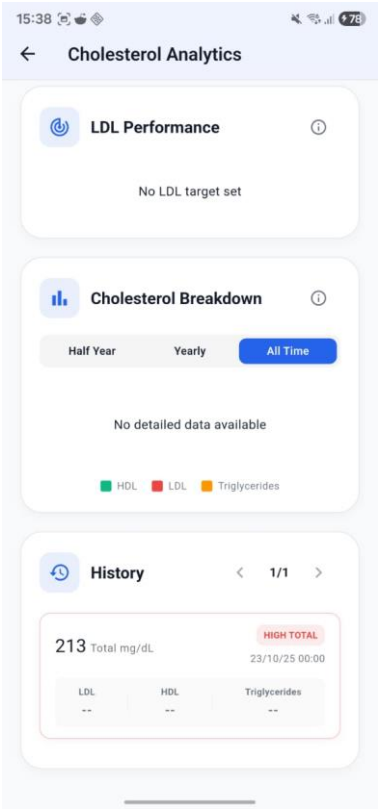
Patient HbA1c (Blood Glucose Level) Analytics



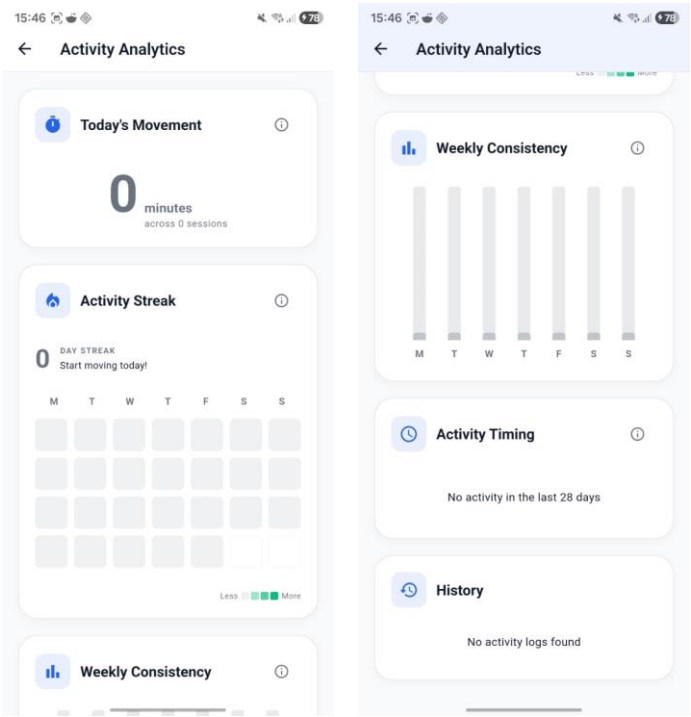


Patient Cholesterol Analytics

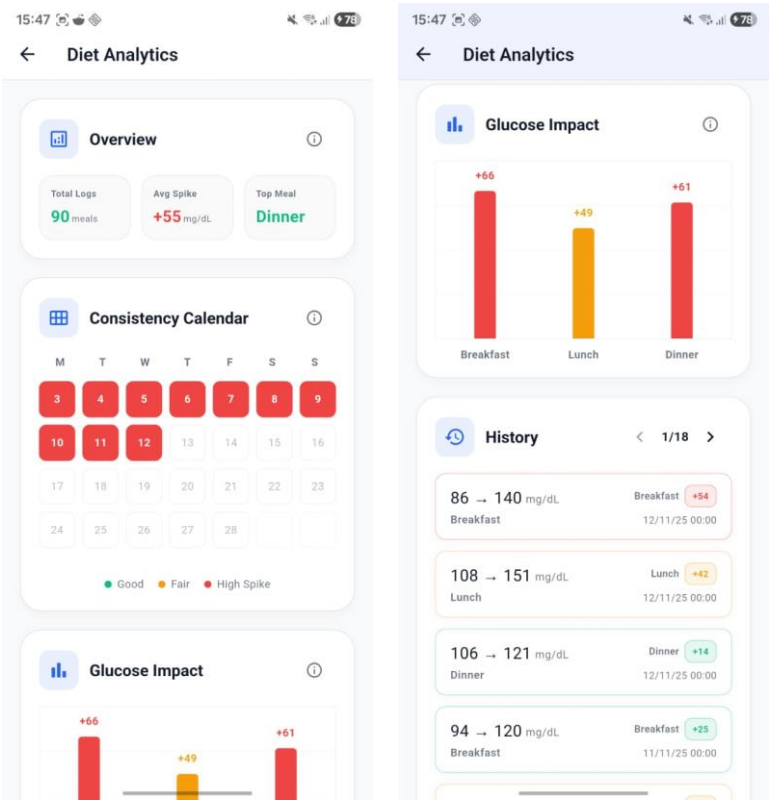




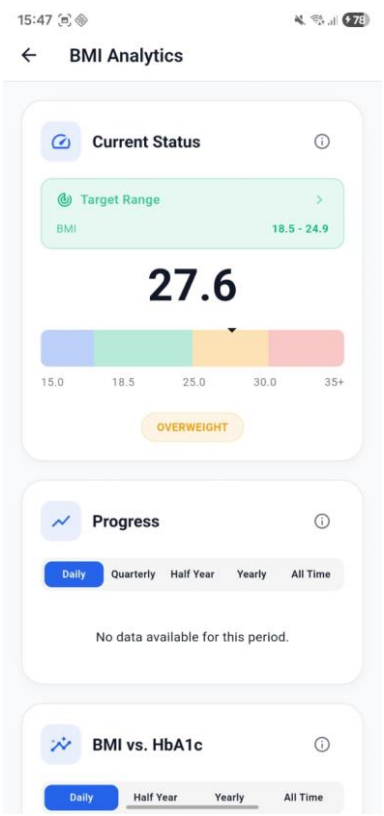
Patient Activity Analytics

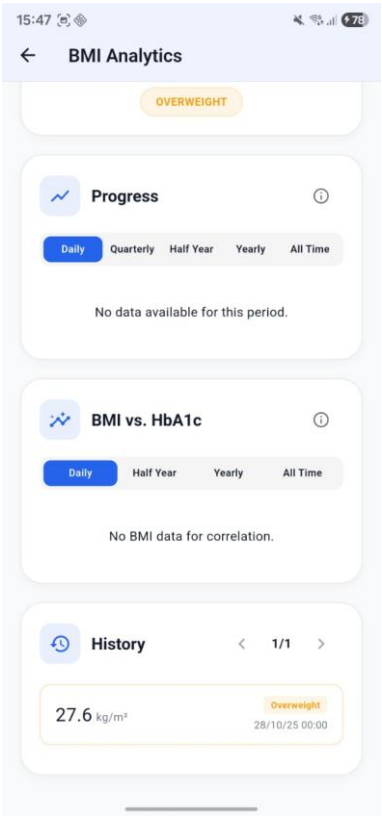


Patient Diet Analytics



Patient BMI Analytics

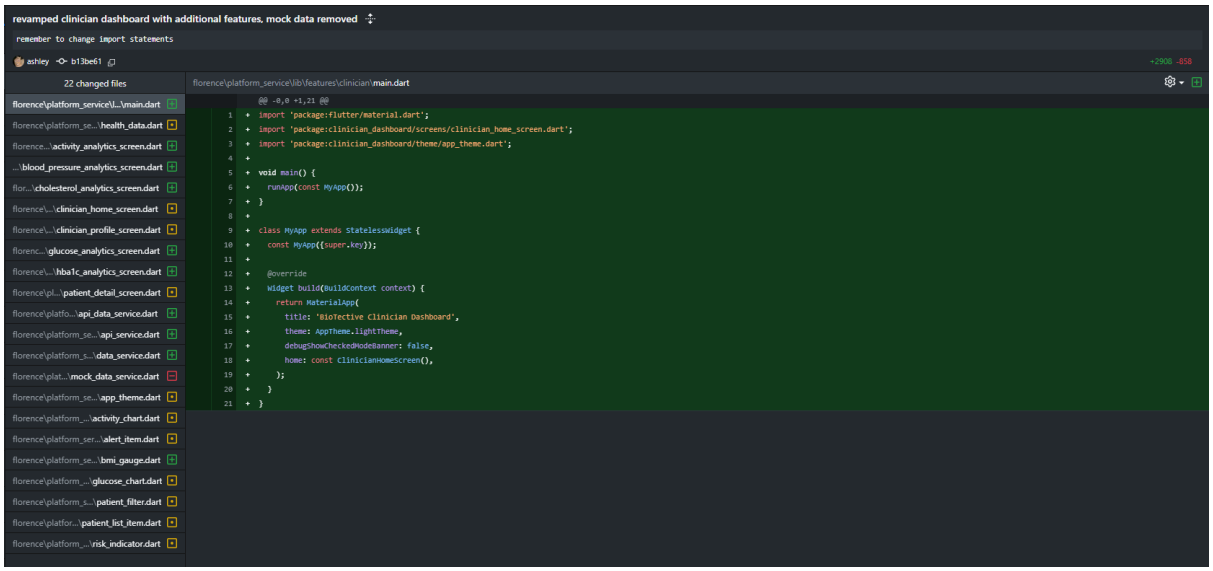




2.3. Ashley (Frontend Developer - Clinician Dashboard)

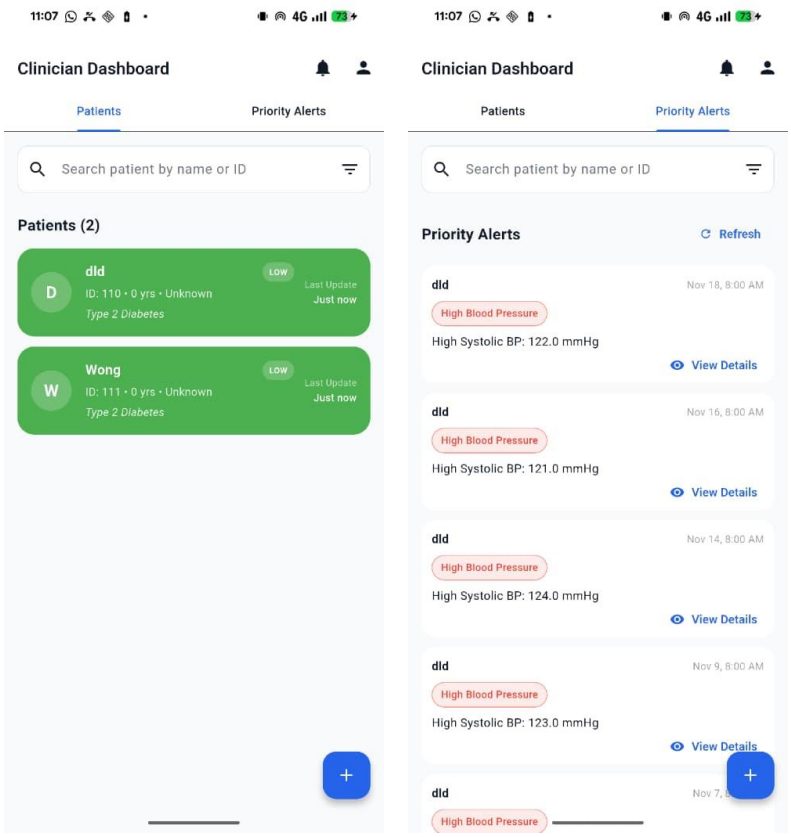
Evidence of Work

GitHub Commits

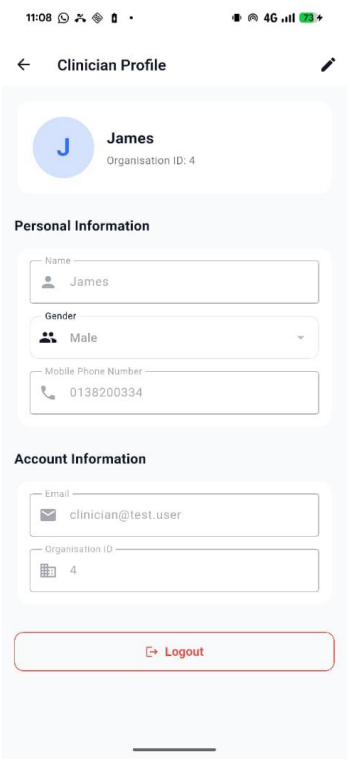


Screenshot of Application

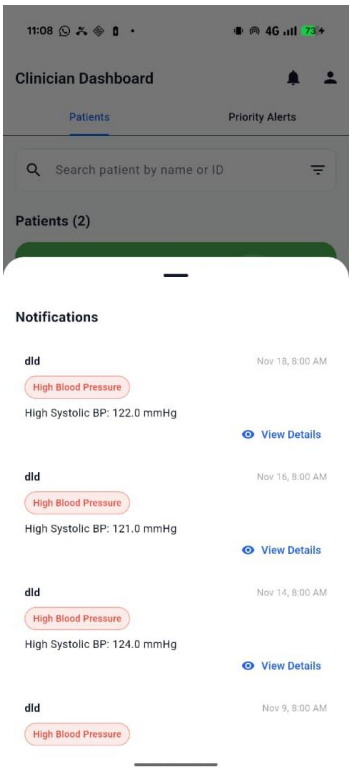
Clinician Homepage



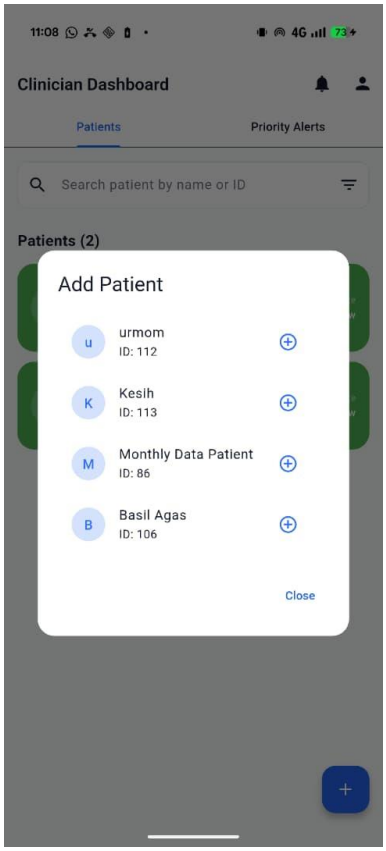
Clinician Profile Page



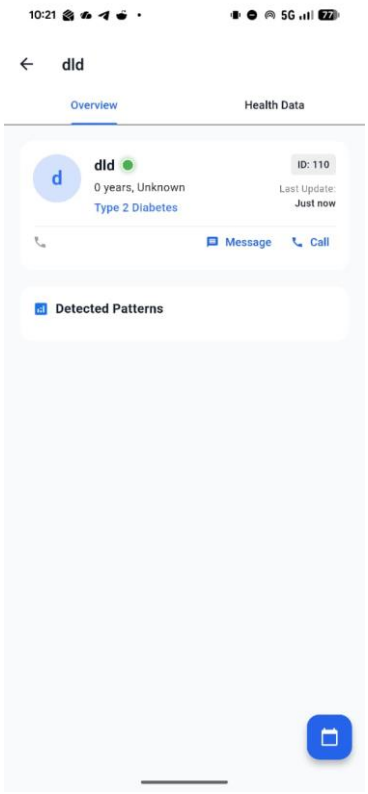
Clinician Notification



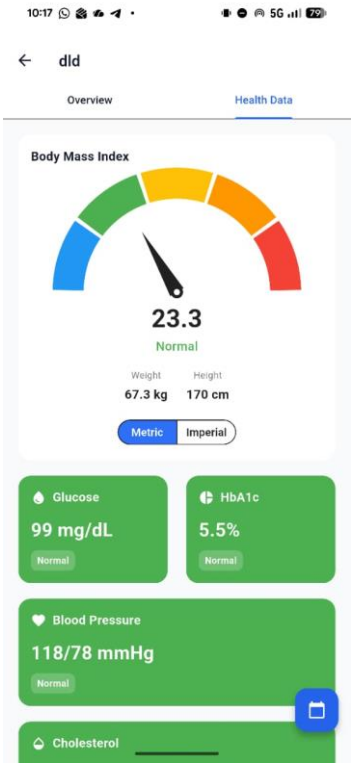
Clinician Adding Patients

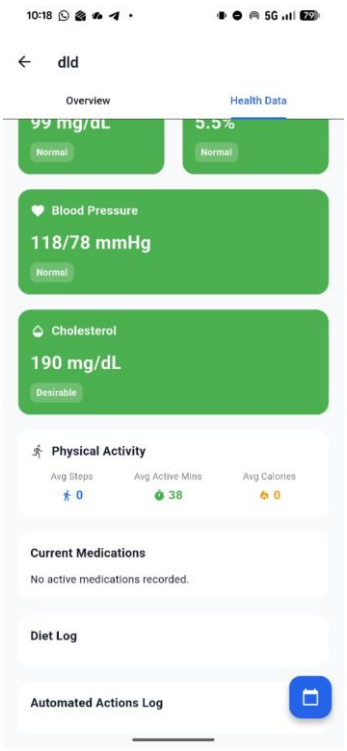


Patient’s Profile Overview

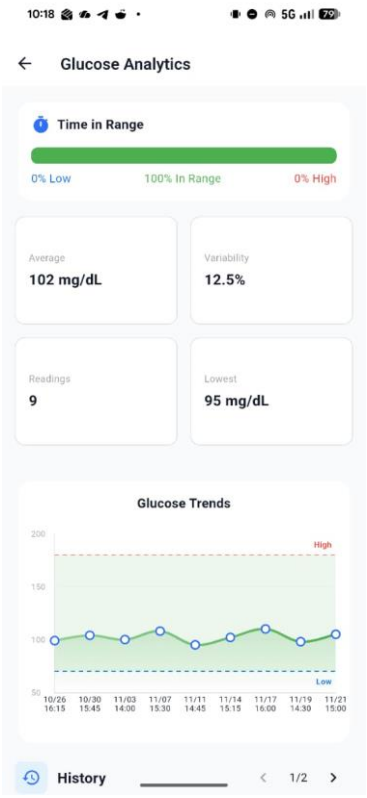


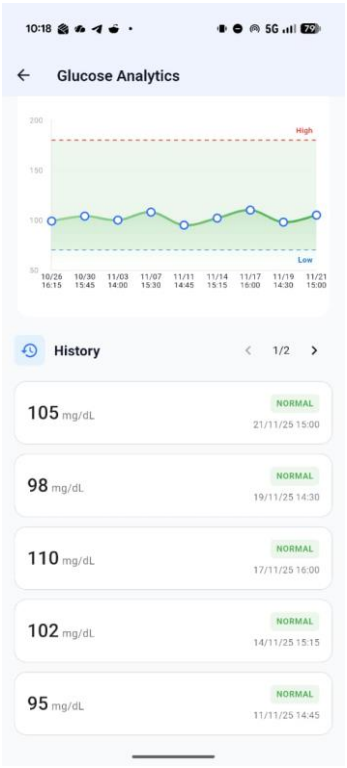
Patient’s Health Data



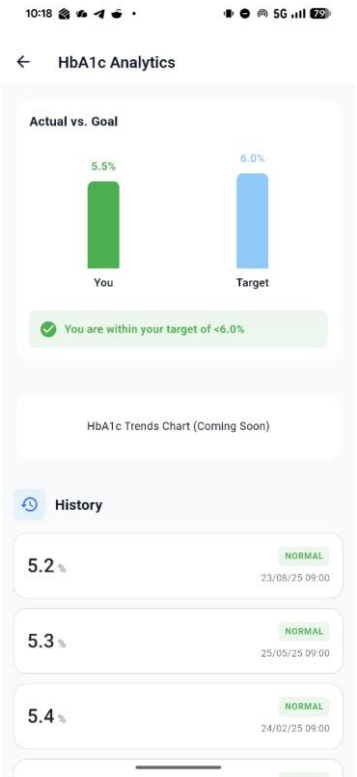


Patient’s Glucose Analytics

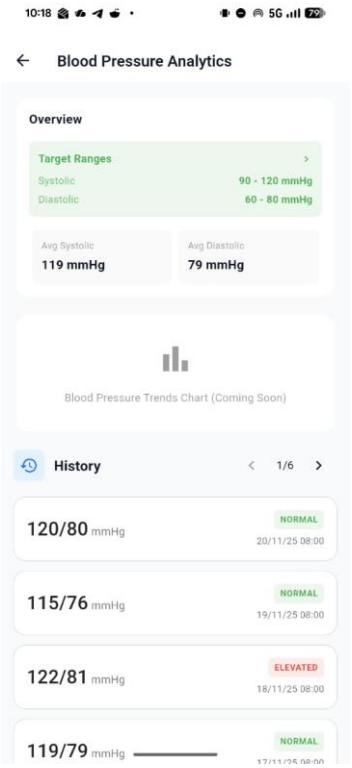




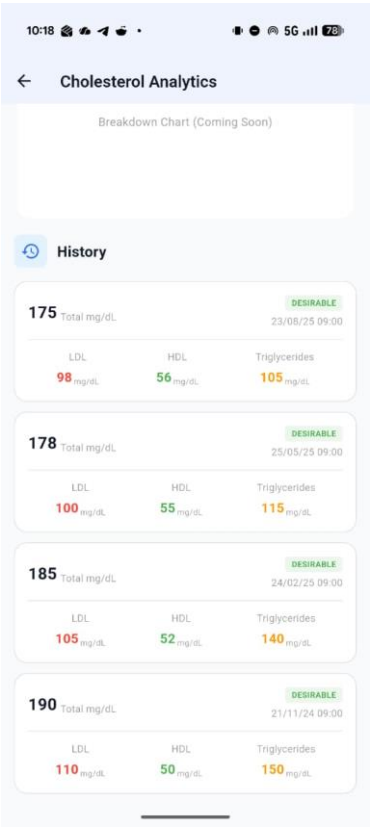
Patient’s HbA1c’s Analytics



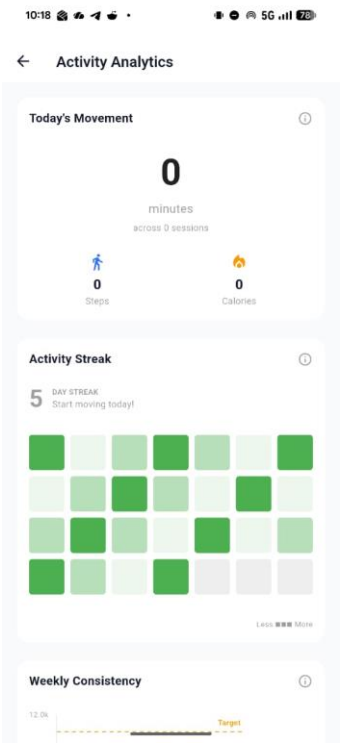
Patient’s Blood Pressure Analytics

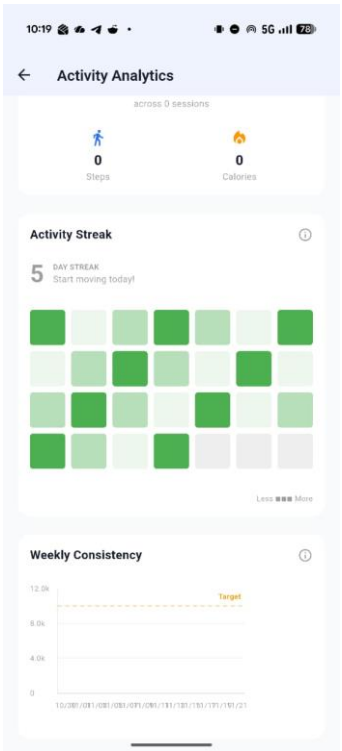


Patient’s Cholesterol Analytics



Patient’s Activity Analytics





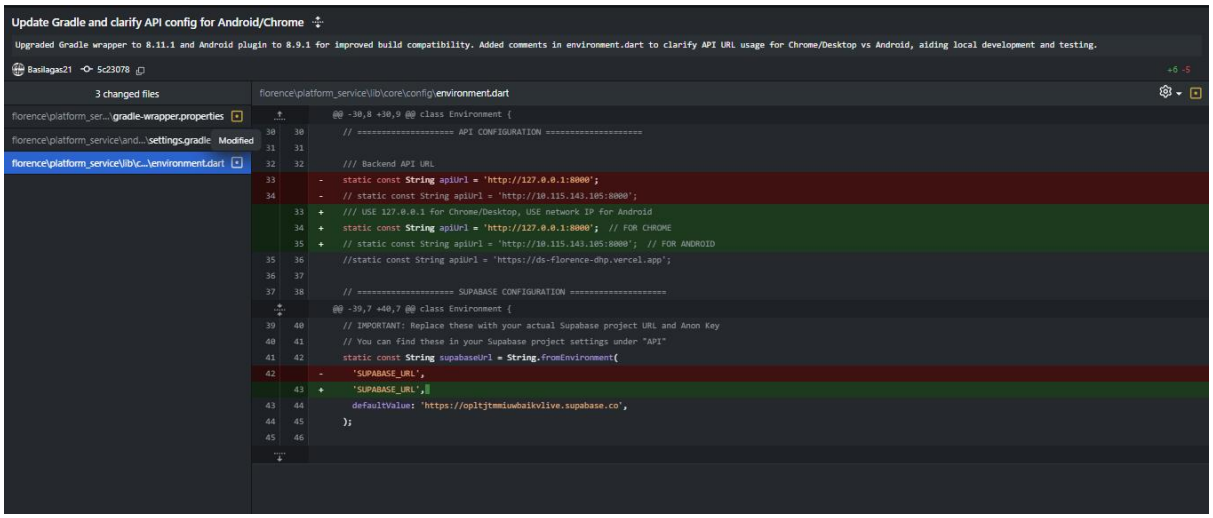
2.4. Basil (Lead AI Engineer and Data Architect)

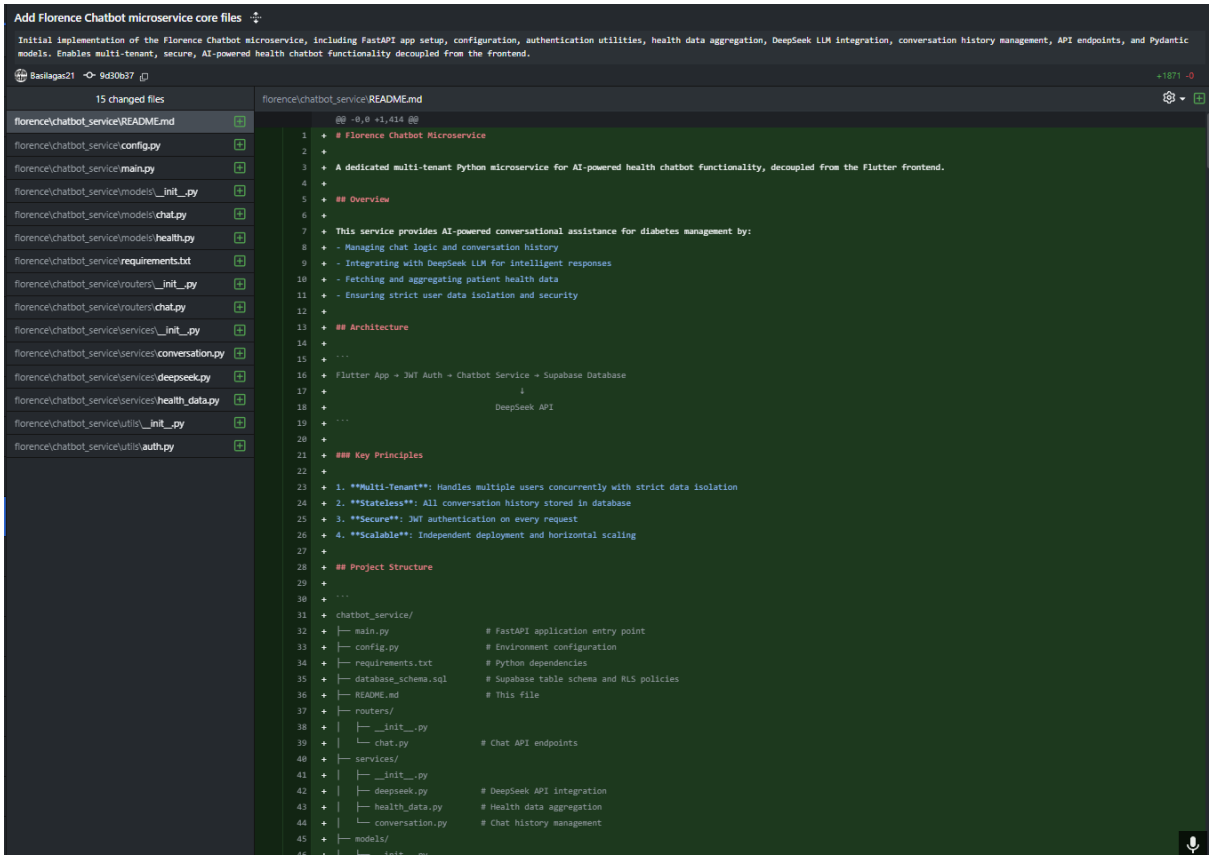
Planned: Connect the AI functionality such as the chatbot, AI powered Recommendation, and the AI powered of the app to the working backend. Also, populate the database with patient's data

Actual: Due to the refactor of the design and 50% of the functionality of the app, I needed to make a fallback measurement and temporarily remove the AI powered Reccomendation and the Pattern Recognition. Therefore, I was able to maintain the functionality of the AI chatbot where the API is securely and privately kept the API key from the Github repository. The Chatbot also now able to store the conversation of each of the patients and the patient user can choose to delete the conversation.

Evidence of Work

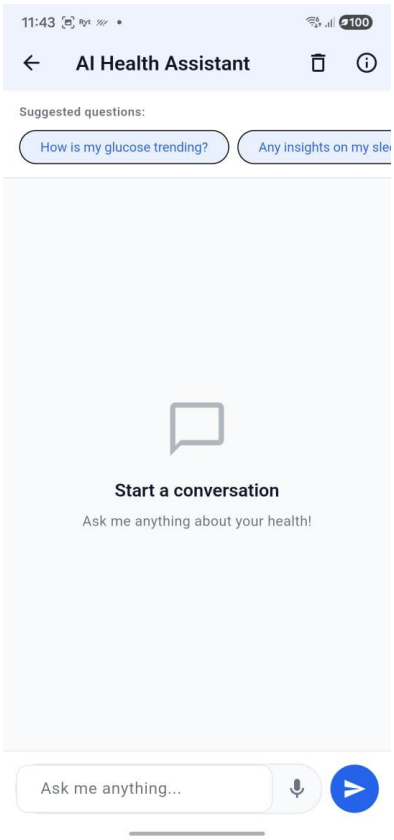
Github commits



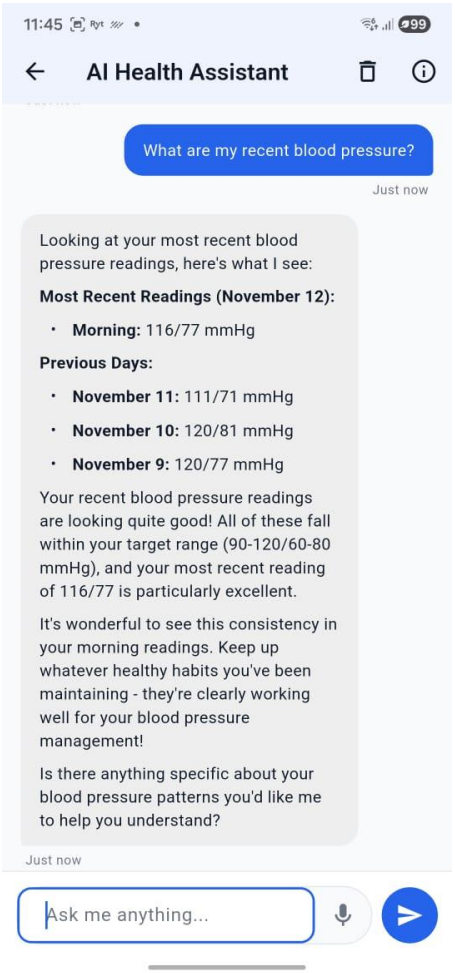

Add Florence Chatbot microservice core files
Initial implementation of the Florence Chatbot microservice, including FastAPI app setup, configuration, authentication utilities, health data aggregation, DeepSeek LLM integration, conversation history management, API endpoints, and Pydantic models. Enables multi-tenant, secure, AI-powered health chatbot functionality decoupled from the frontend.
Basilagas21
9d30b37
+1871 -0
15 changed files
florence\chatbot_service\README.md
florence\chatbot_service\config.py
florence\chatbot_service\main.py
florence\chatbot_service\models__init__.py
florence\chatbot_service\models\chat.py
florence\chatbot_service\models\health.py
florence\chatbot_service\requirements.txt
florence\chatbot_service\routers__init__.py
florence\chatbot_service\routers\chat.py
florence\chatbot_service\services__init__.py
florence\chatbot_service\services\conversation.py
florence\chatbot_service\services\deepseek.py
florence\chatbot_service\services\health_data.py
florence\chatbot_service\utils__init__.py
florence\chatbot_service\utils\auth.py
florence\chatbot_service\README.md
@@ -0,0 +1,414 @@
# Florence Chatbot Microservice
A dedicated multi-tenant Python microservice for AI-powered health chatbot functionality, decoupled from the Flutter frontend.
Overview
This service provides AI-powered conversational assistance for diabetes management by:
- Managing chat logic and conversation history
- Integrating with DeepSeek LLM for intelligent responses
- Fetching and aggregating patient health data
- Ensuring strict user data isolation and security
Architecture
Flutter App → JWT Auth → Chatbot Service → Supabase Database
↓
DeepSeek API
Key Principles
1. **Multi-Tenant**: Handles multiple users concurrently with strict data isolation
2. **Stateless**: All conversation history stored in database
3. **Secure**: JWT authentication on every request
4. **Scalable**: Independent deployment and horizontal scaling
Project Structure
chatbot_service/
├── main.py # FastAPI application entry point
├── config.py # Environment configuration
├── requirements.txt # Python dependencies
├── database_schema.sql # Supabase table schema and RLS policies
├── README.md # This file
├── routers/
│ ├── __init__.py
│ └── chat.py # Chat API endpoints
├── services/
│ ├── __init__.py
│ ├── deepseek.py # DeepSeek API integration
│ ├── health_data.py # Health data aggregation
│ └── conversation.py # Chat history management
├── models/

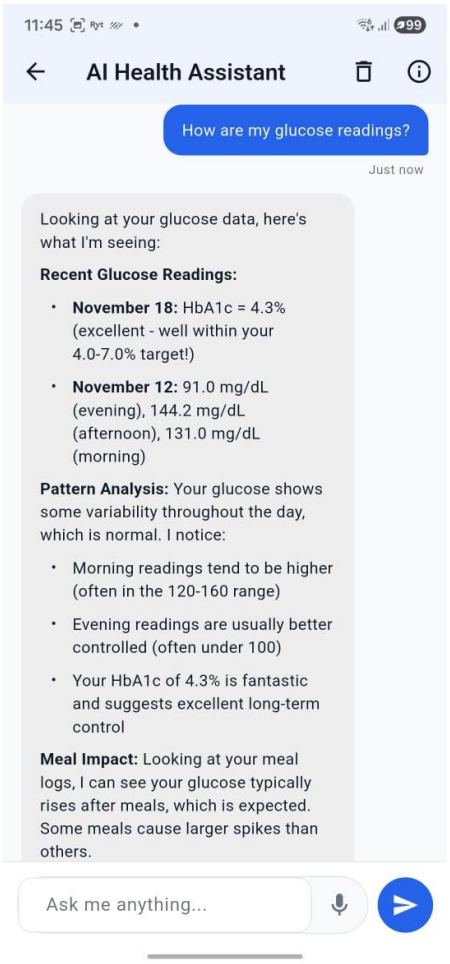
Screenshots of the Application

AI Chatbot UI Page (Default)

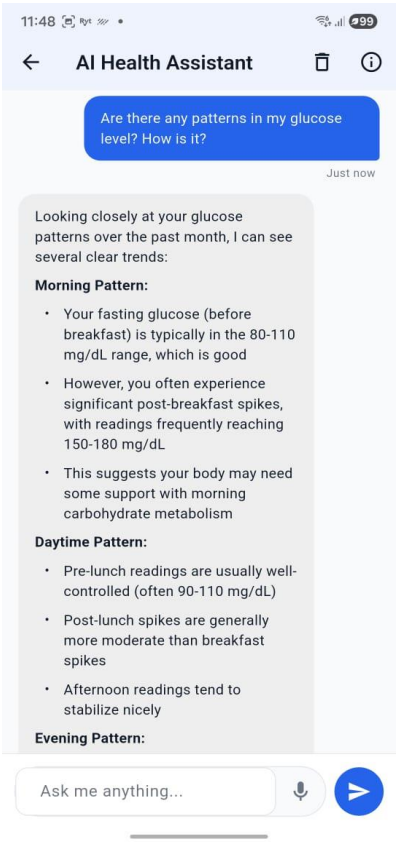


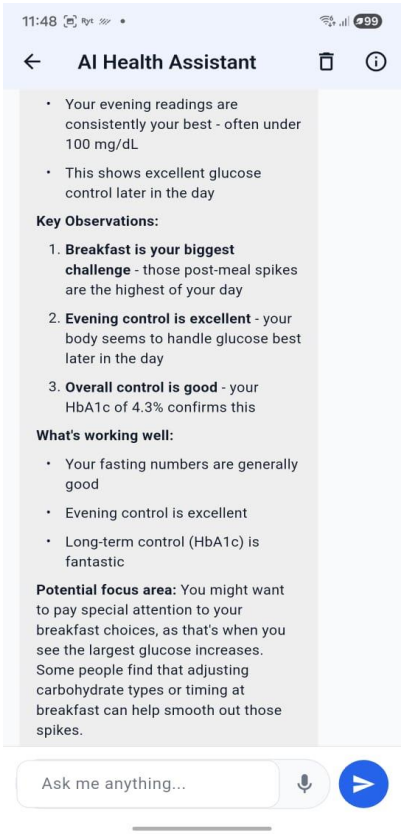
AI Chatbot Test (Health Data)

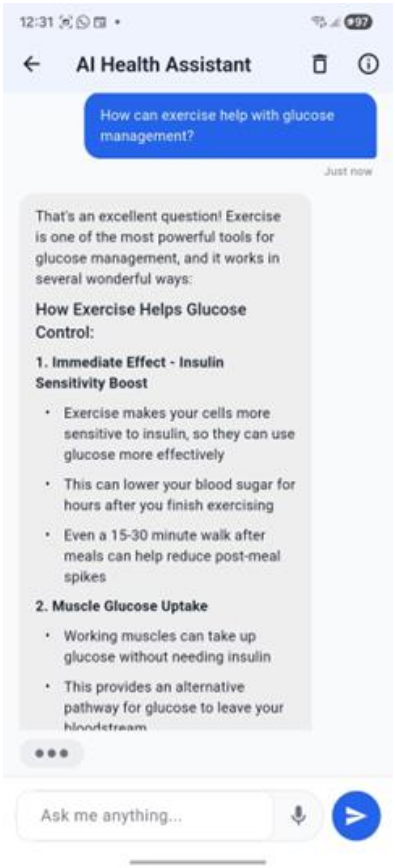


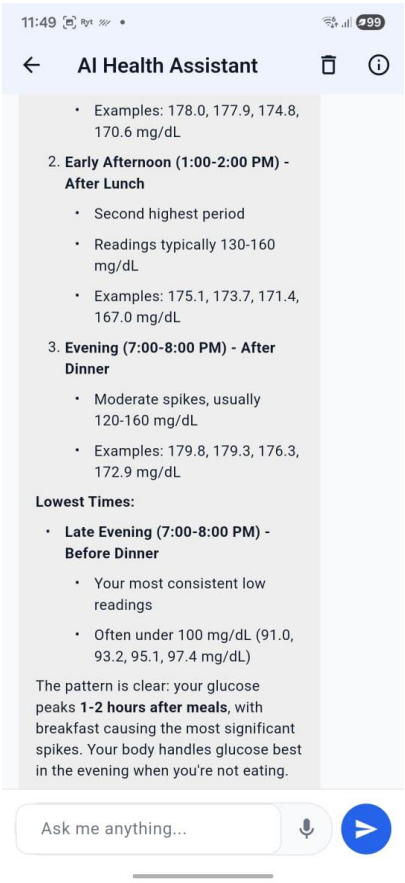


AI Chatbot Test (Patterns and Trends)

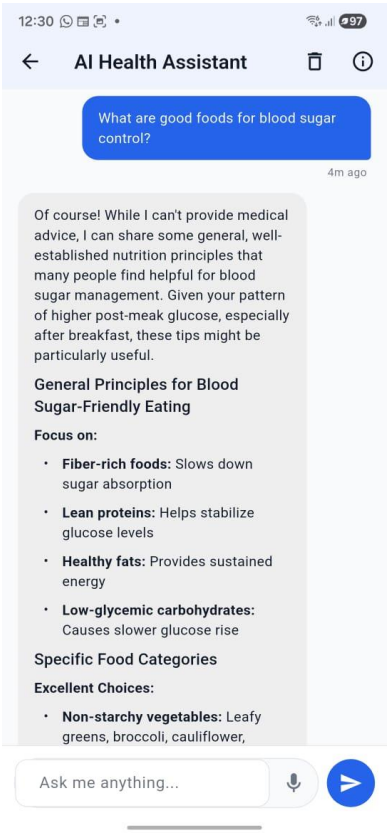


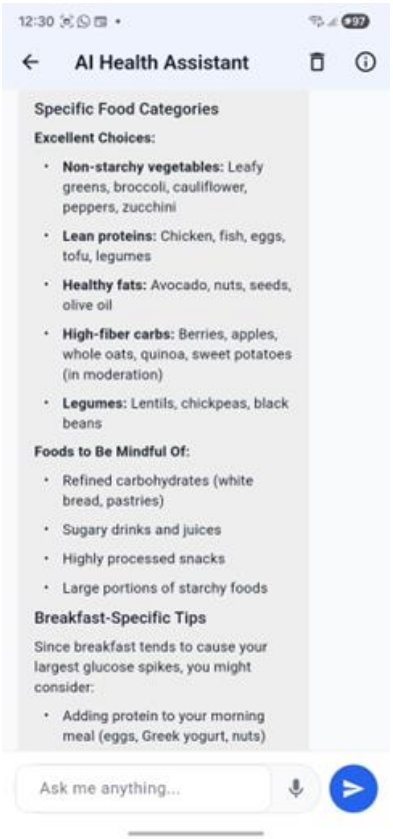


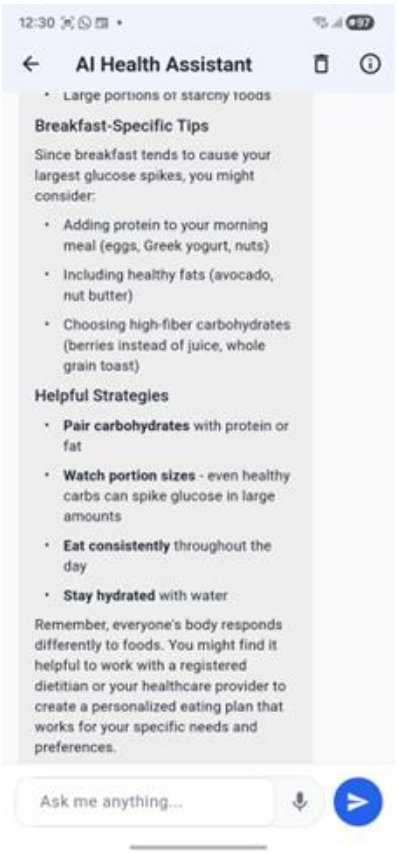


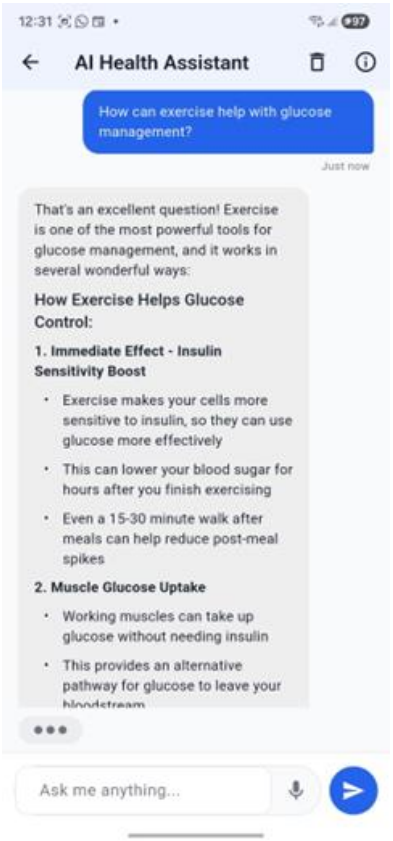


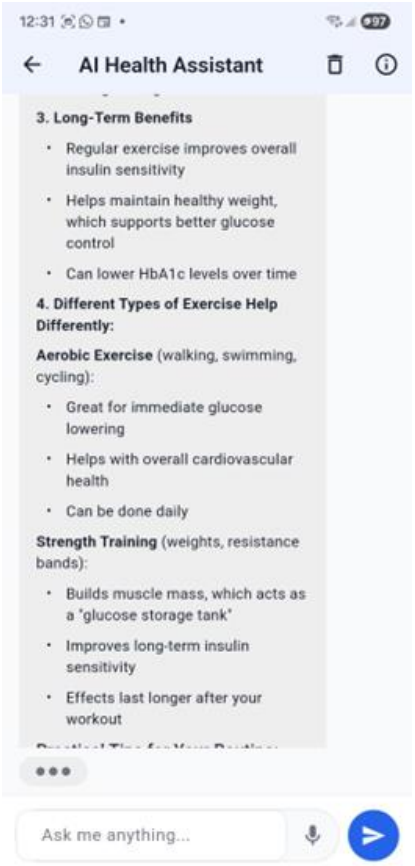
AI Chatbot Test (General Health Questions)

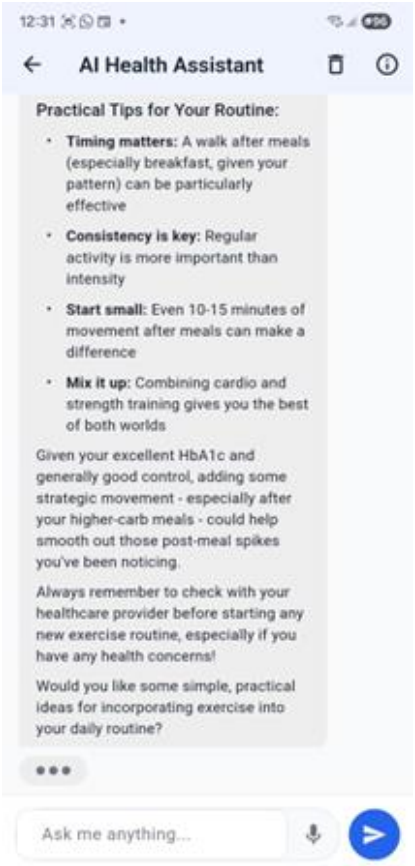


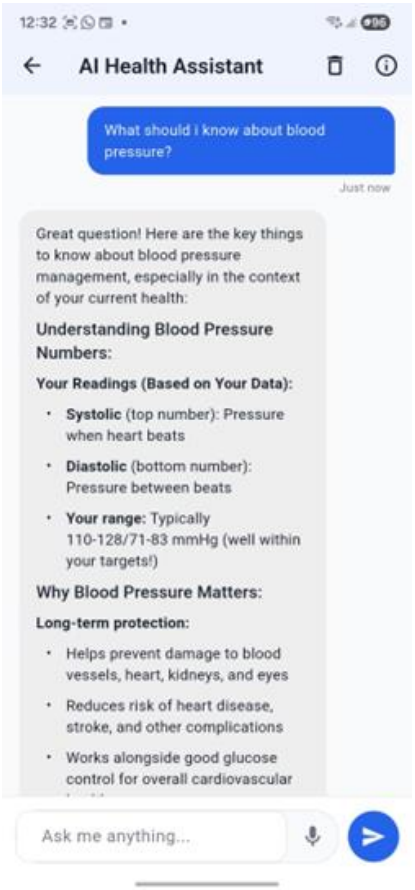




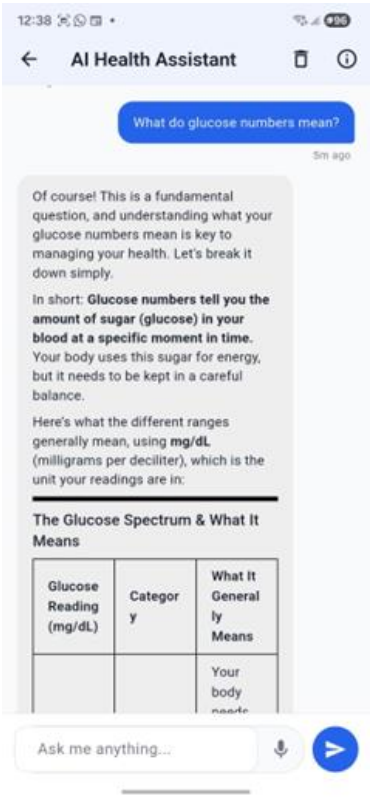


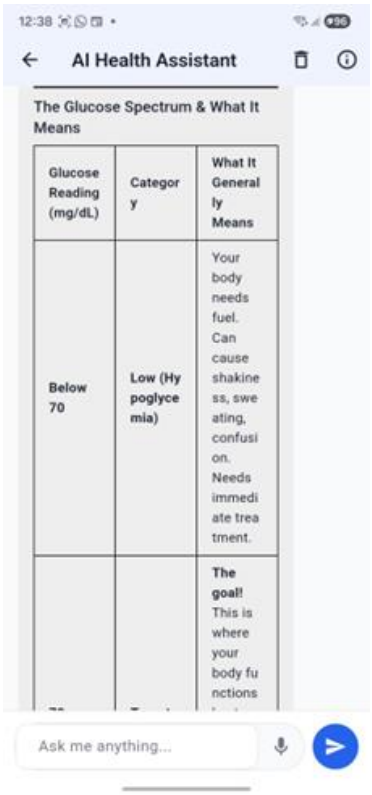

12:31
AI Health Assistant
3. Long-Term Benefits
Regular exercise improves overall insulin sensitivity
Helps maintain healthy weight, which supports better glucose control
Can lower HbA1c levels over time
4. Different Types of Exercise Help Differently:
Aerobic Exercise (walking, swimming, cycling):
Great for immediate glucose lowering
Helps with overall cardiovascular health
Can be done daily
Strength Training (weights, resistance bands):
Builds muscle mass, which acts as a "glucose storage tank"
Improves long-term insulin sensitivity
Effects last longer after your workout
Ask me anything...

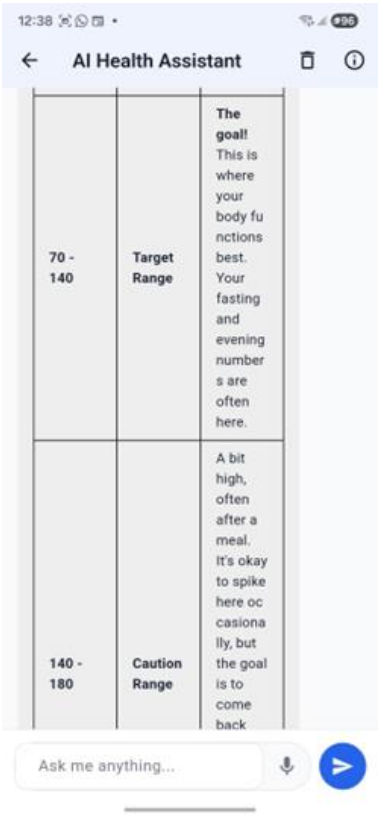

12:31
AI Health Assistant
Practical Tips for Your Routine:
Timing matters: A walk after meals (especially breakfast, given your pattern) can be particularly effective
Consistency is key: Regular activity is more important than intensity
Start small: Even 10-15 minutes of movement after meals can make a difference
Mix it up: Combining cardio and strength training gives you the best of both worlds
Given your excellent HbA1c and generally good control, adding some strategic movement - especially after your higher-carb meals - could help smooth out those post-meal spikes you've been noticing.
Always remember to check with your healthcare provider before starting any new exercise routine, especially if you have any health concerns!
Would you like some simple, practical ideas for incorporating exercise into your daily routine?
Ask me anything...

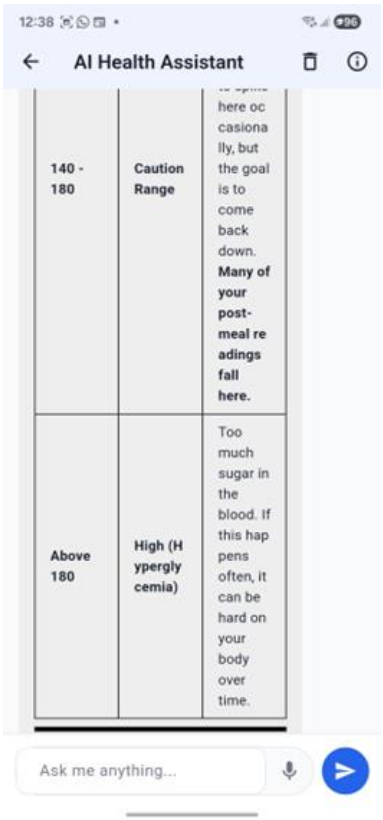

12:32
AI Health Assistant
What should i know about blood pressure?
Just now
Great question! Here are the key things to know about blood pressure management, especially in the context of your current health:
Understanding Blood Pressure Numbers:
Your Readings (Based on Your Data):
Systolic (top number): Pressure when heart beats
Diastolic (bottom number): Pressure between beats
Your range: Typically 110-128/71-83 mmHg (well within your targets!)
Why Blood Pressure Matters:
Long-term protection:
Helps prevent damage to blood vessels, heart, kidneys, and eyes
Reduces risk of heart disease, stroke, and other complications
Works alongside good glucose control for overall cardiovascular
Ask me anything...

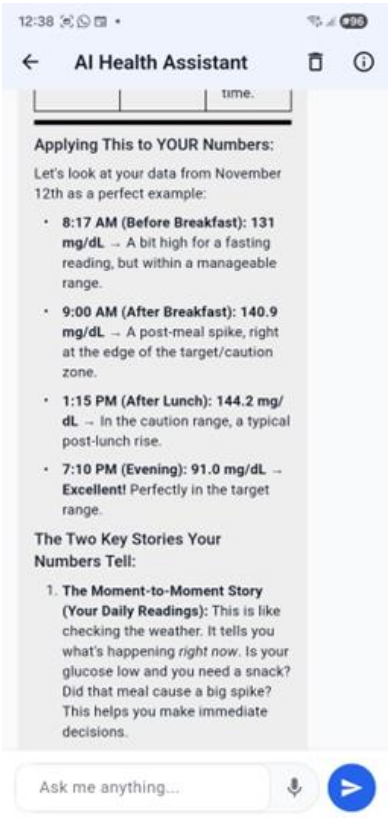
AI Chatbot Test (Understanding the metrics/number/measurement)

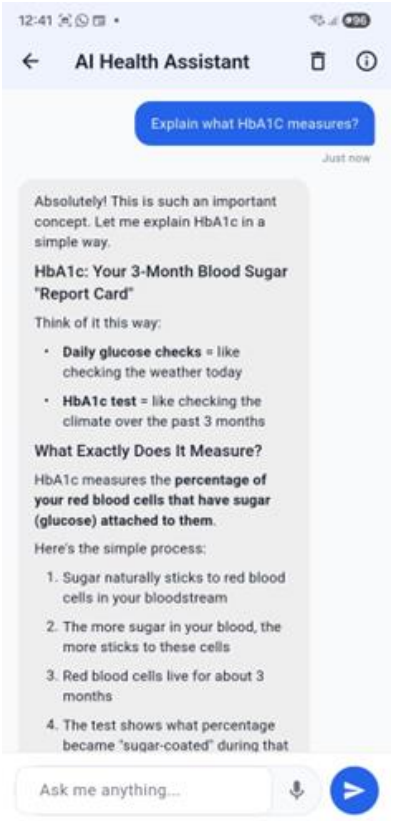


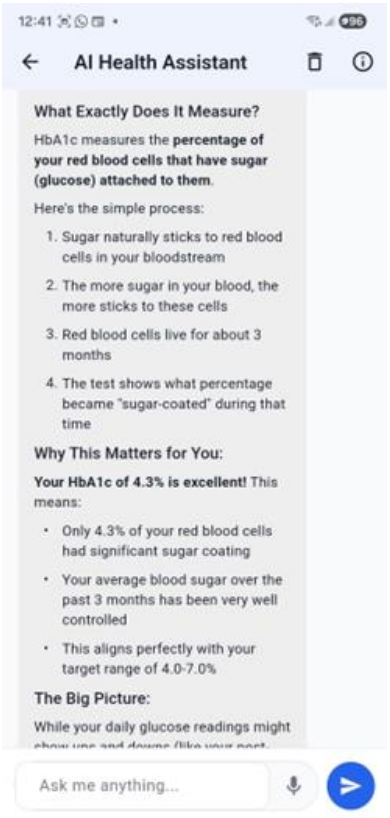


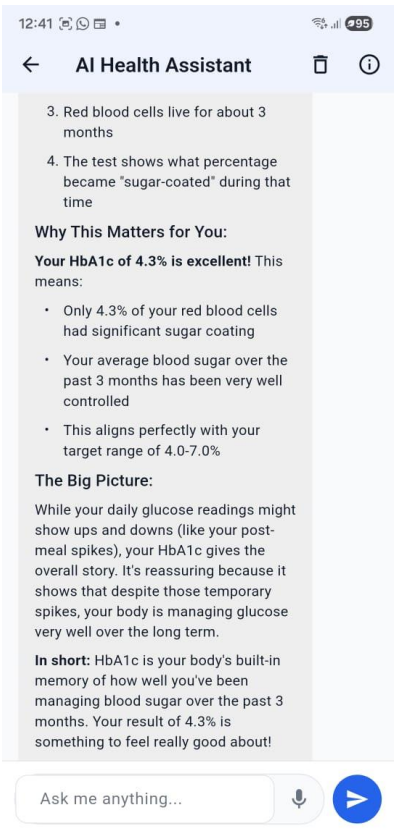




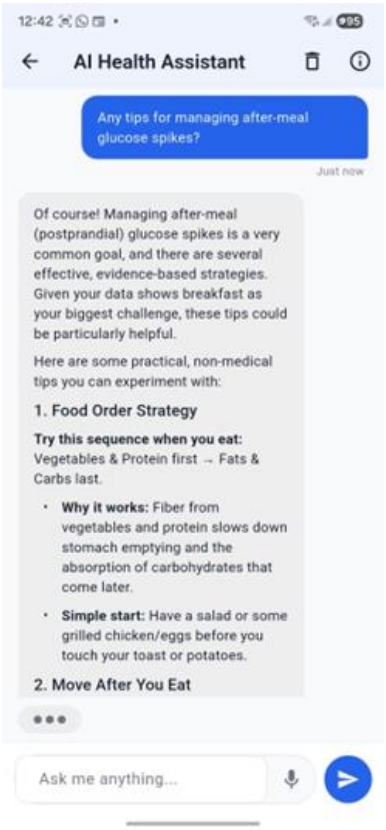

12:38
AI Health Assistant
time.
Applying This to YOUR Numbers:
Let's look at your data from November 12th as a perfect example:
8:17 AM (Before Breakfast): 131 mg/dL → A bit high for a fasting reading, but within a manageable range.
9:00 AM (After Breakfast): 140.9 mg/dL → A post-meal spike, right at the edge of the target/caution zone.
1:15 PM (After Lunch): 144.2 mg/dL → In the caution range, a typical post-lunch rise.
7:10 PM (Evening): 91.0 mg/dL → Excellent! Perfectly in the target range.
The Two Key Stories Your Numbers Tell:
1. The Moment-to-Moment Story (Your Daily Readings): This is like checking the weather. It tells you what's happening right now. Is your glucose low and you need a snack? Did that meal cause a big spike? This helps you make immediate decisions.
Ask me anything...

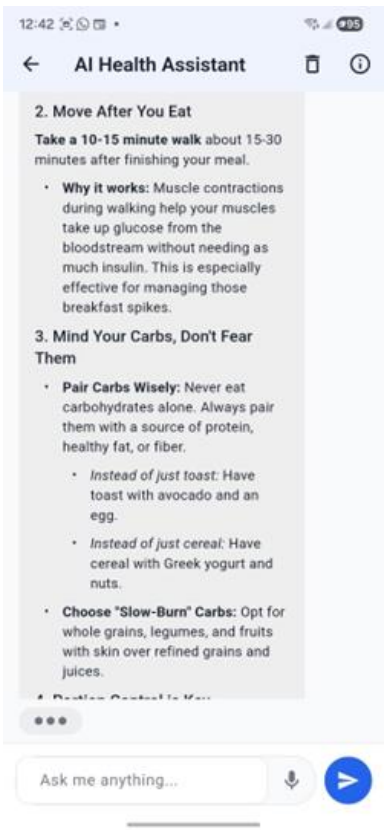

12:41
AI Health Assistant
Explain what HbA1C measures?
Just now
Absolutely! This is such an important concept. Let me explain HbA1c in a simple way.
HbA1c: Your 3-Month Blood Sugar "Report Card"
Think of it this way:
Daily glucose checks = like checking the weather today
HbA1c test = like checking the climate over the past 3 months
What Exactly Does It Measure?
HbA1c measures the percentage of your red blood cells that have sugar (glucose) attached to them.
Here's the simple process:
1. Sugar naturally sticks to red blood cells in your bloodstream
2. The more sugar in your blood, the more sticks to these cells
3. Red blood cells live for about 3 months
4. The test shows what percentage became "sugar-coated" during that
Ask me anything...

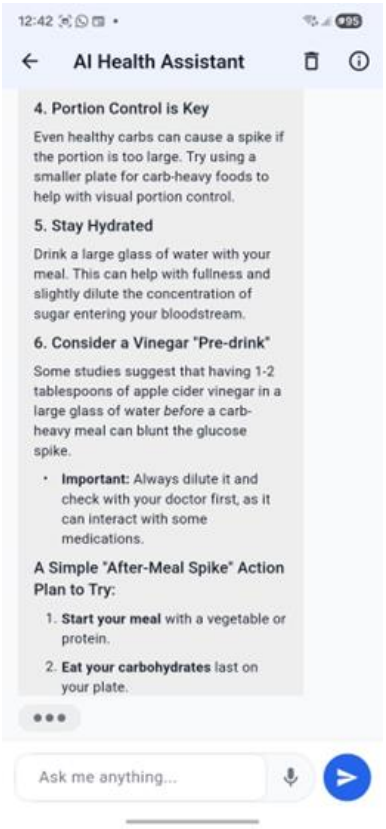

12:41
AI Health Assistant
What Exactly Does It Measure?
HbA1c measures the percentage of your red blood cells that have sugar (glucose) attached to them.
Here's the simple process:
1. Sugar naturally sticks to red blood cells in your bloodstream
2. The more sugar in your blood, the more sticks to these cells
3. Red blood cells live for about 3 months
4. The test shows what percentage became "sugar-coated" during that time
Why This Matters for You:
Your HbA1c of 4.3% is excellent! This means:
Only 4.3% of your red blood cells had significant sugar coating
Your average blood sugar over the past 3 months has been very well controlled
This aligns perfectly with your target range of 4.0-7.0%
The Big Picture:
While your daily glucose readings might
Ask me anything...

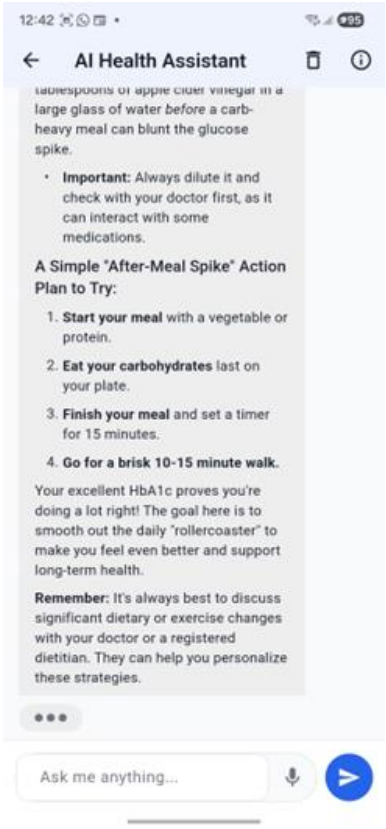

12:41
AI Health Assistant
3. Red blood cells live for about 3 months
4. The test shows what percentage became "sugar-coated" during that time
Why This Matters for You:
Your HbA1c of 4.3% is excellent! This means:
Only 4.3% of your red blood cells had significant sugar coating
Your average blood sugar over the past 3 months has been very well controlled
This aligns perfectly with your target range of 4.0-7.0%
The Big Picture:
While your daily glucose readings might show ups and downs (like your post-meal spikes), your HbA1c gives the overall story. It's reassuring because it shows that despite those temporary spikes, your body is managing glucose very well over the long term.
In short: HbA1c is your body's built-in memory of how well you've been managing blood sugar over the past 3 months. Your result of 4.3% is something to feel really good about!
Ask me anything...

AI Chatbot Test (Daily Management)\

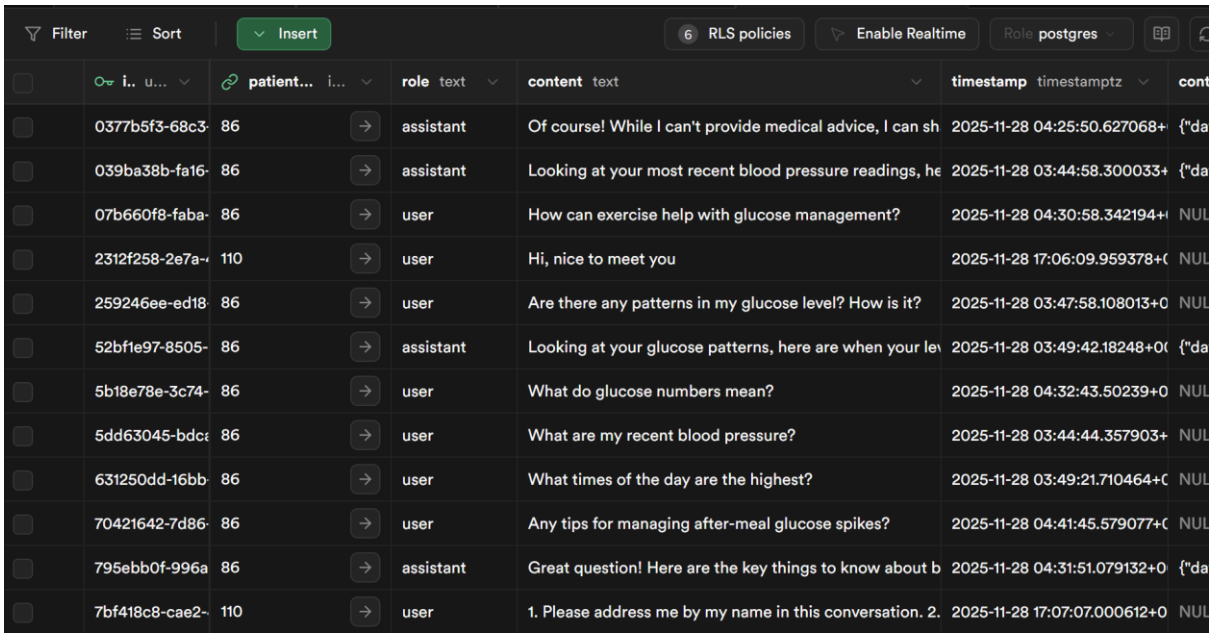








AI Chatbot with the Working Backend



Filter | Sort | Insert | 6 RLS policies | Enable Realtime | Role postgres

	i.. u...	patient... i...	role text	content text	timestamp timestamptz	cont
	0377b5f3-68c3-	86	assistant	Of course! While I can't provide medical advice, I can sh	2025-11-28 04:25:50.627068+	{"da
	039ba38b-fa16-	86	assistant	Looking at your most recent blood pressure readings, he	2025-11-28 03:44:58.300033+	{"da
	07b660f8-faba-	86	user	How can exercise help with glucose management?	2025-11-28 04:30:58.342194+(	NUL
	2312f258-2e7a-(	110	user	Hi, nice to meet you	2025-11-28 17:06:09.959378+(	NUL
	259246ee-ed18-	86	user	Are there any patterns in my glucose level? How is it?	2025-11-28 03:47:58.108013+0	NUL
	52bf1e97-8505-	86	assistant	Looking at your glucose patterns, here are when your lev	2025-11-28 03:49:42.18248+0(	{"da
	5b18e78e-3c74-	86	user	What do glucose numbers mean?	2025-11-28 04:32:43.50239+0	NUL
	5dd63045-bdca	86	user	What are my recent blood pressure?	2025-11-28 03:44:44.357903+	NUL
	631250dd-16bb-	86	user	What times of the day are the highest?	2025-11-28 03:49:21.710464+(	NUL
	70421642-7d86-	86	user	Any tips for managing after-meal glucose spikes?	2025-11-28 04:41:45.579077+(	NUL
	795ebb0f-996a	86	assistant	Great question! Here are the key things to know about b	2025-11-28 04:31:51.079132+0	{"da
	7bf418c8-cae2-(	110	user	1. Please address me by my name in this conversation. 2.	2025-11-28 17:07:07.000612+0	NUL

2.5. Edison (Backend Developer)

My contribution this sprint bridged the gap between advanced backend AI logic and the frontend user experience. I worked in close collaboration with **Daniel** to design and implement the Data Visualisation dashboard.

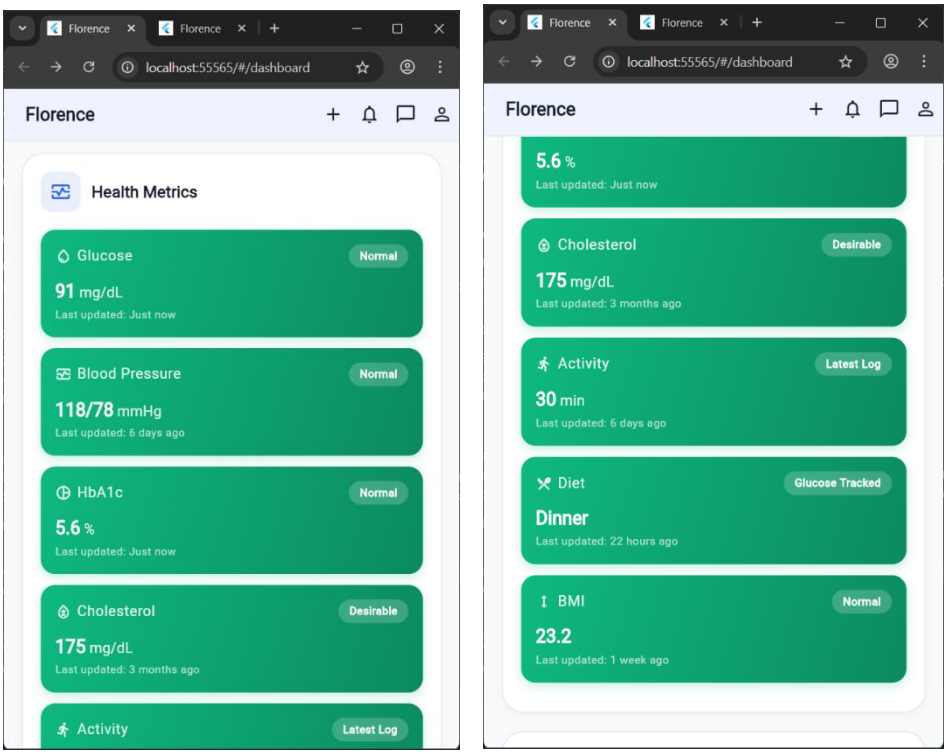
Frontend Data Visualisation:

On the client side, I worked alongside **Daniel** to implement the comprehensive **Health Dashboard and Data Visualisation suite**.

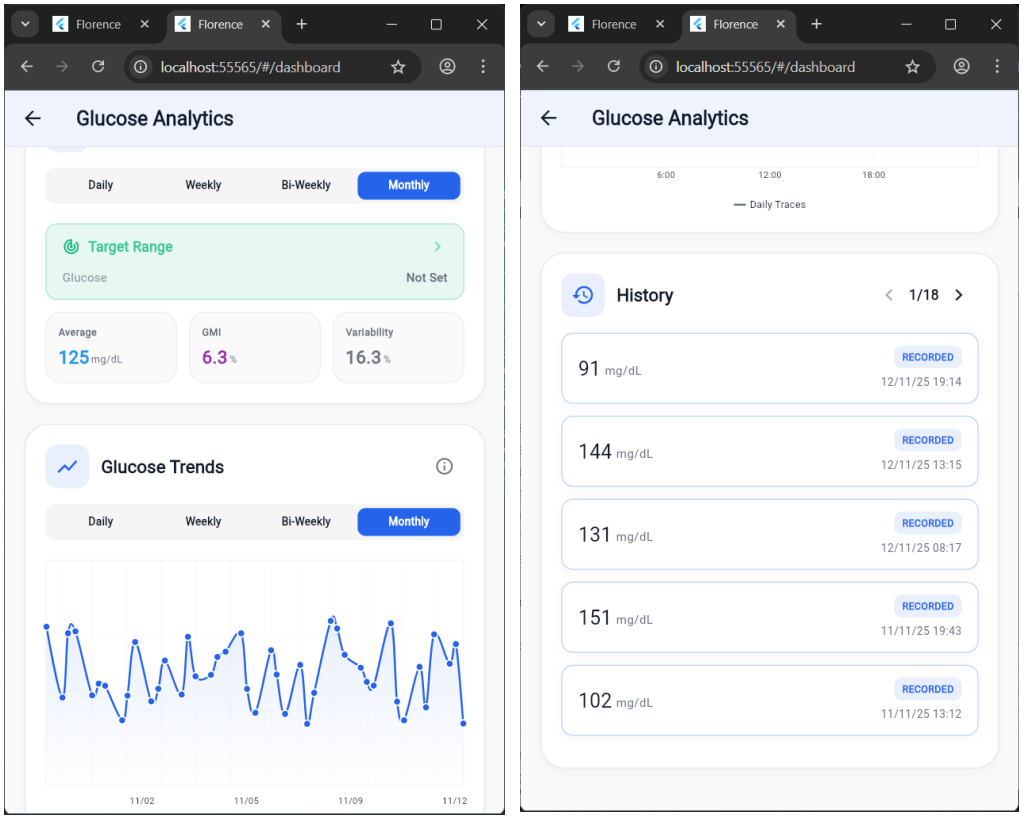
- **Interactive Analytics:** We co-developed the detailed analytics screens for Glucose, Blood Pressure, and HbA1c. I utilised fl_chart to build complex visualisations, such as dynamic line charts for trends and scatter plots for correlation analysis.
- **Biometric Cards:** I implemented the reusable CompactHealthCard and BiometricsSection widgets to summarise critical health metrics, while collaborating with **Daniel** to ensure the state management (via Riverpod) correctly updated these UI components in real-time.

Evidence of Work

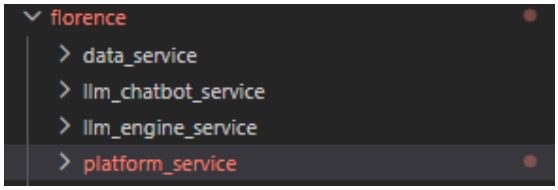
The “Biometrics” Dashboard (Frontend)



Deep Dive Analytics Chart (Frontend)



The Microservice Architecture (File Structure)



Context Injection Logic (Code)

```python
# Singleton instance
_health_data_service: Optional[HealthDataService] = None


def get_health_data_service() -> HealthDataService:
    """Get or create the singleton HealthDataService instance."""
    global _health_data_service
    if _health_data_service is None:
        _health_data_service = HealthDataService()
    return _health_data_service

```

3. Sprint Plan

This sprint focused on **EPIC 3: Integration and Intelligence**, covering Milestone 5 and parts of 6. Following the architectural refactoring in Sprint 2, the primary goal was to operationalise the Data Service as the central source of truth and fully implement the patient-facing AI Chatbot.

A critical strategic decision was made during Sprint Planning to **prioritise the conversational Chatbot and Data Visualisation** over the background Automation Engine (LAM Triggers). This decision was based on the client's preference for a tangible, interactive user experience for the prototype demonstration.

Sprint Backlog

Story ID	User Story	Priority	Story Points	Status
T3.1	As a patient, I want to chat with an AI assistant that remembers my previous context so that I can have a natural conversation about my health.	High	13	Completed
T2.2	As a patient, I want to see my detailed Glucose, BP and BMI data from the backend displayed as interactive charts populated with real data from the backend.	High	8	Completed
T1.3	As a user, I want a robust authentication system (Email/Password) with deep-linking support for password resets and verification.	High	8	Completed
T5.1	As a clinician, I want my dashboard to fetch real patient lists and see priority patients.	High	13	Completed
T6.1	As a developer, I want to implement a dedicated Microservice for the Chatbot	Medium	8	Completed

	to handle LLM context and state management separately from the core data API.			
T4.1	*[DEFERRED]* As a developer, I want to implement the background LLM Automation Engine for proactive triggers.	Low	13	Moved to Next Semester

Table 1: Sprint 3 Backlog showing completed core features and the strategic deferral of the automation engine.

4. Sprint Progress

The team achieved significant velocity this sprint, successfully integrating the frontend applications with the backend services. The system has moved from a "mock-up" phase to a fully functional "connected" prototype.

4.1. The Data Service (Central Hub)

We successfully finalised the **Florence Data Service** (FastAPI). It now acts as the single source of truth, handling all CRUD operations for patients, clinicians and health logs.

I. **Security and Access Control**: We implemented a comprehensive Row-Level Security (RLS) architecture on Supabase to enforce strict data isolation across three distinct user roles:
   a. **Patients:** Policies utilise *auth.uid()* to ensure patients have exclusive Read/Write access only to records directly linked to their own *user_id* (e.g., *patient_monitor_data*, *patient_chat_history*).
   b. **Clinicians:** Implemented relational policies that grant read access to patient data only if the *patient_profiles.clinician_id* matches the authenticated clinician's ID. This ensures that clinicians can only view data for patients who are explicitly assigned to them.
   c. **Admins:** Implemented Role-Based Access Control (RBAC), examining the JWT *app_metadata*. Users with the *ADMIN* role claim are granted global management permissions across all tables (profiles, logs and organisations) for system oversight.

II. **API Development:** Developed a rich suite of RESTful endpoints organised by functional domain to support distinct user workflows:
   a. **Patient Self-Management**: Implemented "me"-scoped endpoints allowing patients to log and retrieve diverse health metrics (Glucose, BP, HbA1c, BMI), track lifestyle data (Meals, Activity) and view their personalised health thresholds.
   b. **Clinician Monitoring:** Created aggregation endpoints that allow clinicians to fetch their assigned patient roster, view detailed patient histories, access computed priority alerts and append clinical notes.

c. **System Administration:** Developed high-level management endpoints for onboarding users, configuring global settings and managing patient-clinician assignments.

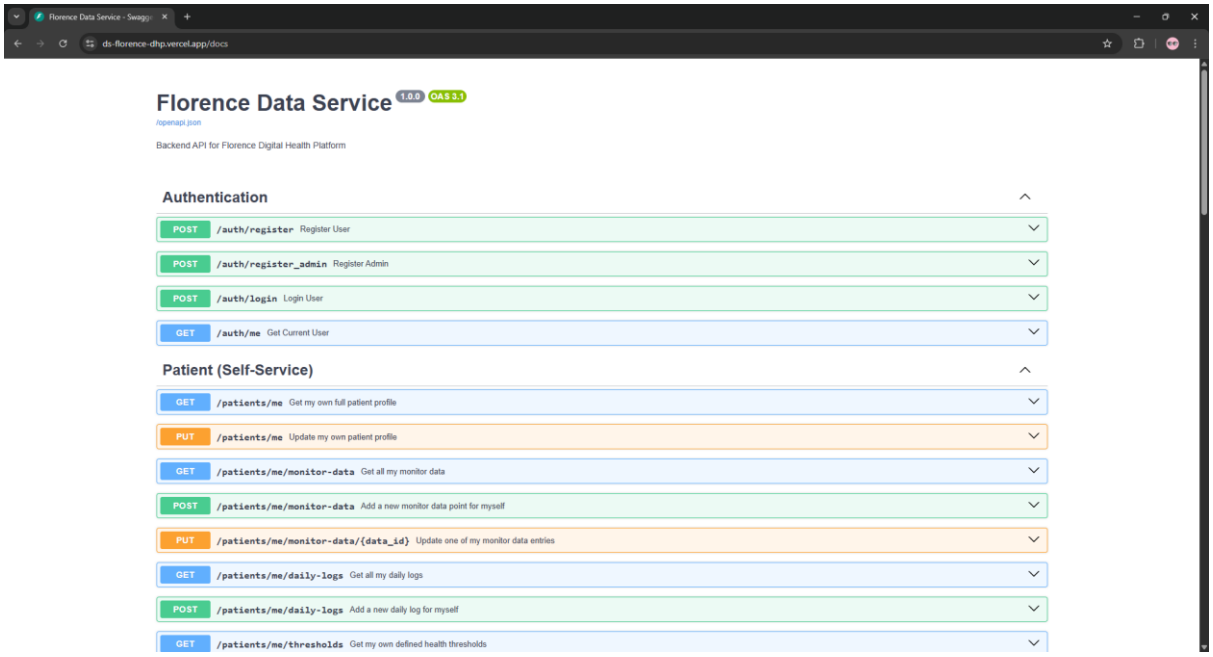


Figure 1: Swagger UI/OpenAPI documentation showing the full list of active endpoints

4.2. Patient Dashboard and Visualisation

The Patient App underwent a major architectural and visual overhaul to improve usability and data clarity.

I. **Structural Refinement**: We redesigned the dashboard to enforce a clear separation of concerns. AI-driven features were consolidated into a dedicated "AI Insight Card", while standard vital signs were organised into a distinct "Health Metrics" section. This allows users to instinctively differentiate between raw health data and AI-generated guidance.

II. **Live Data Integration:** Replaced static placeholders with the *fl_chart* library, fully integrated with our *HealthDataProvider*. Charts now render real-time historical data fetched directly from the Data Service.

III. **Dynamic Visual Logic:** Implemented threshold-aware rendering logic. Visual elements automatically shift colors (Green/Amber/Red) based on the patient's specific health targets (e.g., a glucose reading > 180 mg/dL instantly turns the data point red), providing immediate, at-a-glance health context.

IV. **Targeted Visualisation Strategy**: We conducted research to select the most appropriate visualisation type for each specific metric (e.g., using Scatter Charts for Blood Pressure correlation vs. Line Charts for Glucose trends), ensuring the analytics pages provide maximum readability and clinical value.

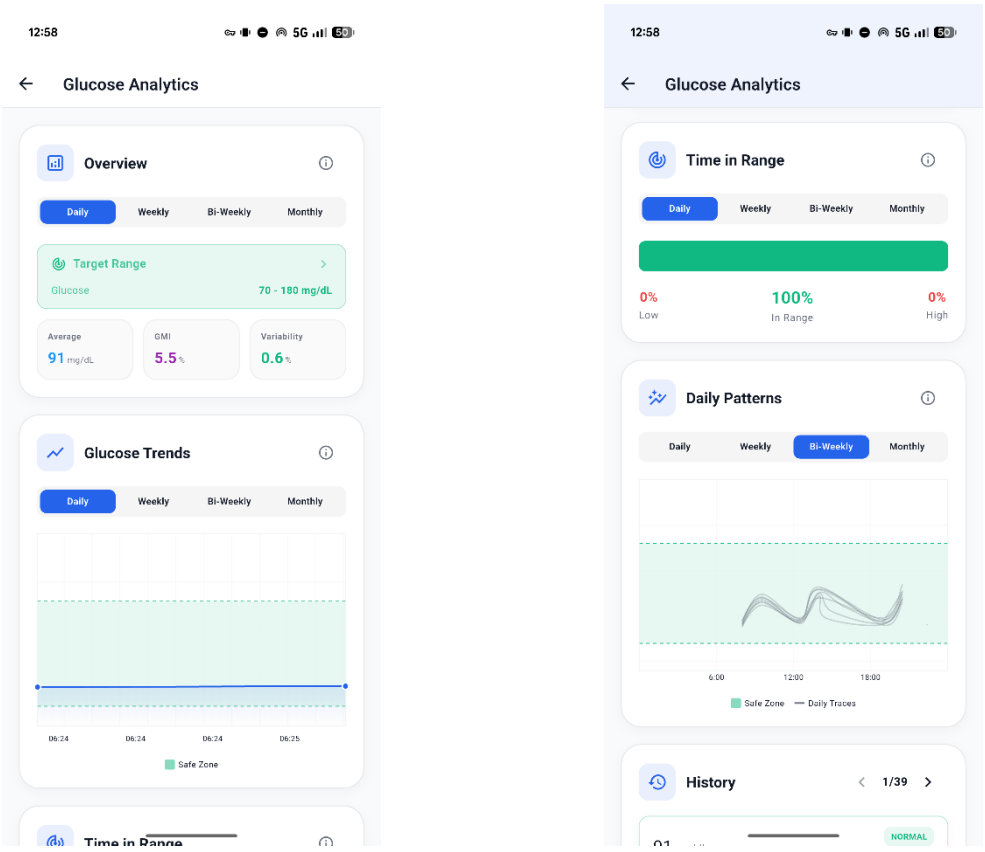


Figure 2: The Glucose Analytics screen showing the graph populated with data points

4.3. AI Chatbot (Context-Aware)

The Chatbot is now fully functional. Unlike a generic wrapper, our implementation includes:

I. **Full-Context Integration:** Leveraging DeepSeek's extensive context window, we architected the system to retrieve and inject the patient's **entire health dataset** (profile, thresholds and historical logs) into the system prompt. This approach allows us to benchmark the model's reasoning capabilities on complete patient histories without the complexity of retrieval algorithms such as (RAG) at this stage.

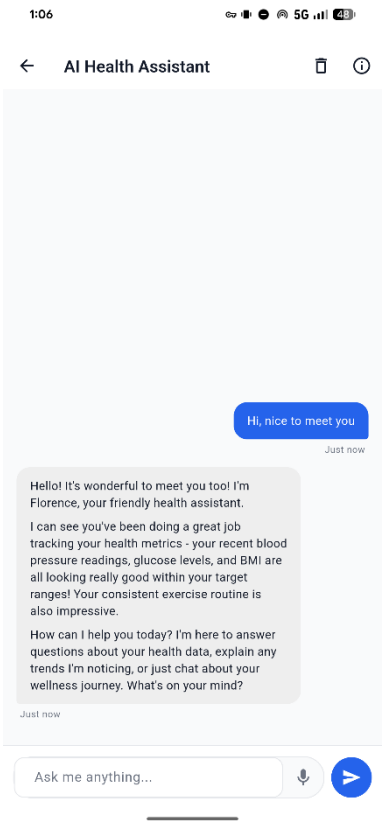
II. **Prompt Engineering:** We rigorously engineered the system prompt to define the "Florence" persona. This involved structuring raw JSON data into a readable format for the LLM and implementing strict behavioural instructions (e.g., "be empathetic," "never diagnose," "reference specific data points"), ensuring the AI remains safe and helpful even with large context inputs.

III. **Architectural Convergence (Tool Calling):** Since the upcoming Automation Engine requires an agentic "Large Action Model" architecture, we plan to refactor the Chatbot to utilise **Tool Calling** (Function Calling) for data retrieval. We will conduct A/B testing to determine if dynamically fetching data via tools offers better performance and lower token usage compared to the current full-context injection method.

IV. **Reliable State Management**: We implemented a persistent conversation state stored in Supabase (*patient_chat_history*), decoupling the chat logic from the UI lifecycle.

a. **Background Processing:** If a user exits the screen while the AI is generating a response, the backend service completes the processing and saves the message to the database independently.

b. **Seamless Synchronisation:** When the user returns, the *ChatbotService* re-syncs the full conversation history. This eliminates common bugs that occur when returning to a chat, resulting in disjointed threads or missing context, and ensures a smooth, continuous user experience even across app restarts.



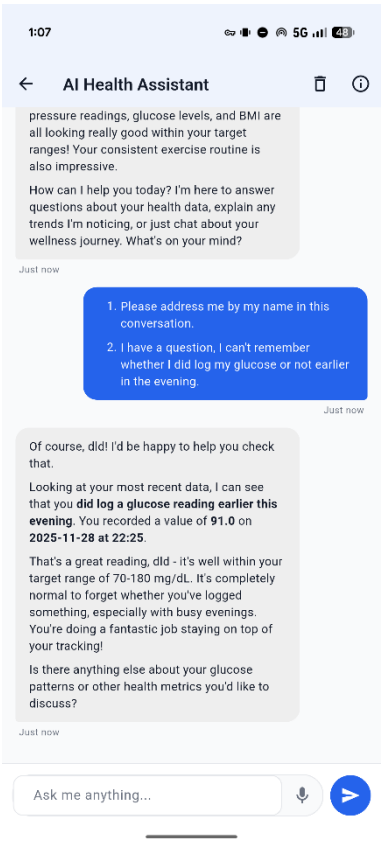


Figure 3: A chat conversation where the AI references the patient's specific glucose reading from the database

4.4. Authentication

We finalised the complete authentication architecture, migrating from development-only anonymous logins to a production-ready system.

I. **Secure Registration:** Enforced email/password authentication using Supabase Auth with strict password policies, ensuring data security for both patients and clinicians.

II. **Frictionless Verification (Deep Linking):** We implemented a seamless onboarding flow using deep links. When a user clicks the verification link in their email, the app launches immediately and automatically establishes an authenticated session. This eliminates the need for manual re-login after verification, significantly reducing user friction during the first run.

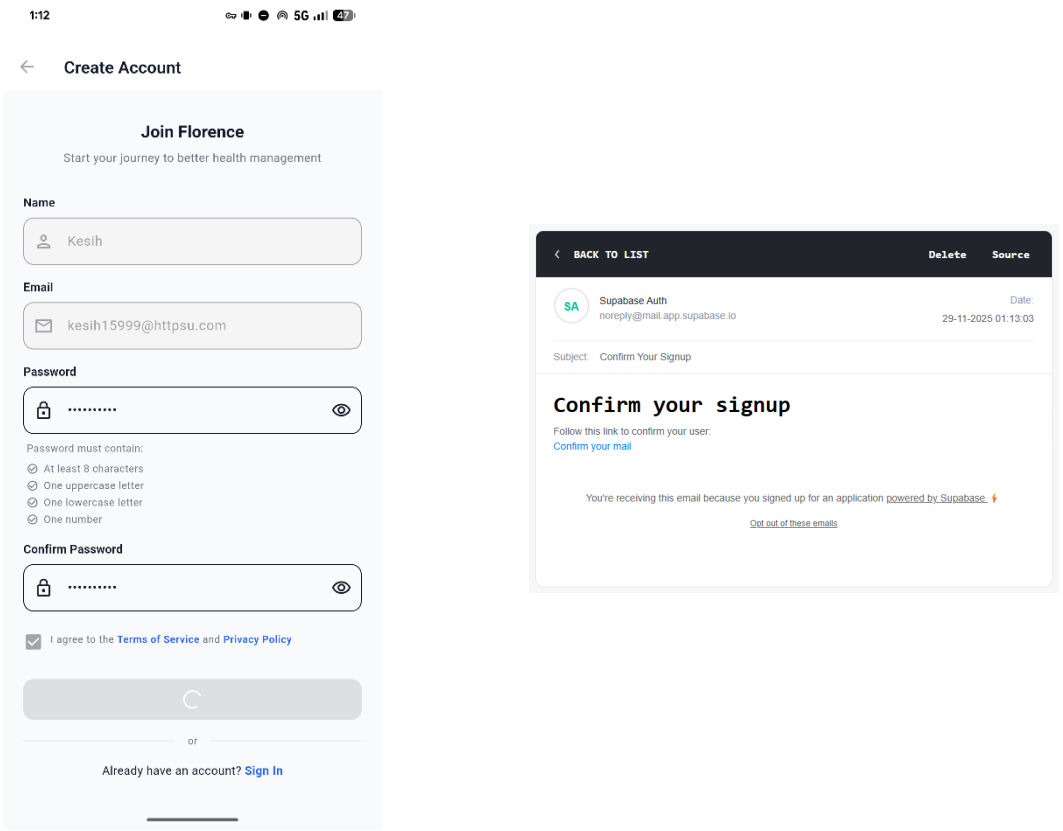


Figure 4: The Register Screen and the "Check your email" verification prompt

5. Sprint Review

The review process for this sprint deviated from the standard live demonstration format due to a strategic pivot directed by the project supervisor during our mid-sprint consultation.

5.1. Consultation and Strategic Realignment

In our consultation mid-sprint, we presented the initial UI mockups and architectural plan. The supervisor provided a clear directive: **prioritise the functional connectivity of the Patient Dashboard over the breadth of features.**

Based on this, the team dedicated the remainder of the sprint to intense development work to finalise the "Vertical Slice" of the Patient App.

I. **Current Status:** The Patient Dashboard is now fully functional, connected to the backend and visually overhauled. Internal testing confirms that data logging, visualisation and the AI Chatbot are operating correctly.
II. **Deployment Strategy**: To overcome the limitations of physical demonstrations, where testing is restricted to the short consultation time slot on a team member's device, we are transitioning to an Asynchronous User Acceptance Testing (UAT) model. Immediately following this submission, we plan to establish a CI/CD pipeline (targeting Firebase App Distribution). This will allow the supervisor to install the functional build on their own device and test it at their convenience, facilitating a much deeper review than a brief in-person session permits.

5.2. Client Feedback

The feedback driving this sprint's output was derived from the mid-sprint consultation:

I. **Directive:** The client explicitly instructed the team to "finish the basic functionality for the patient dashboard first" before expanding to the Clinician side or advanced automation.
II. **Scope Adjustment:** It was mutually agreed to defer the LLM Automation Engine to the next semester. The client emphasised that a solid, working data

platform is more valuable for the final presentation than a buggy experimental feature.

5.3. Critical Analysis

This sprint represents a massive leap in technical maturity, moving the project from "Design" to "Engineering."

I. **Progress Against Plans:** We successfully achieved the sprint's primary goal: delivering a fully connected Patient App. While the Clinician Dashboard backend is ready, we deliberately paused its UX development to ensure the Patient side was polished and bug-free, strictly adhering to the client's quality-first directive.

II. **Challenges and Solutions:**

   a. **Complex State Synchronisation:** Managing state consistency across the Chatbot, Data Logging and Visualisation widgets proved to be the most significant technical hurdle. Initially, data logged in one screen would not immediately reflect in others due to reliance on local widget state. We resolved this by refactoring the entire app to use **Riverpod** for global state management, allowing us to invalidate specific data providers upon successful API writes to trigger instant, synchronised UI updates across the app.

   b. **Clinical Domain Expertise for Visualisation:** We underestimated the domain knowledge required to implement medically useful visualisations. We learned that generic charting libraries are insufficient without customisation, for example, Blood Pressure requires visualising the correlation between Systolic and Diastolic pressure, whereas Glucose data requires highlighting "Time in Range." Bridging the gap between raw data rendering and clinical interpretability required significant research and custom implementation of the *fl_chart* library.

III. **Outcome:** The system is stable. The separation of concerns (Frontend, Data Service, Chatbot Service) allows for independent scaling and easier debugging.

6. Retrospect (Critical Review of The Process)

6.1. Strengths

I. **Microservice Architecture:** Separating the Chatbot logic into its own Python service (FastAPI) proved excellent. It allowed the AI engineers to tweak prompts and context logic without breaking the main mobile app.

II. **Monorepo Strategy:** Continuing the strategy from Sprint 2, the monorepo allowed for easy sharing of data models and rapid iteration between frontend and backend teams.

6.2. Challenges and Bottlenecks

I. **Clinician Workflow Optimisation:** While the Clinician Dashboard successfully integrates with the backend, the User Experience (UX) lacks clinical efficiency. The current "Patient Review" workflow introduces unnecessary friction and navigation steps. Significant iteration is required to streamline the interface, ensuring clinicians can assess patient risk profiles rapidly and intuitively.

II. **Profile Data Complexity:** The patient profile schema proved more complex than anticipated, particularly due to the requirement for granular, personalised health thresholds for every user. Integrating this extensive data model with the frontend forms consumed significant development time. To maintain sprint velocity and ensure stability, we prioritised the display logic over editing capabilities, leaving secondary fields as read-only for this release.

6.3. Ethical and Security Review

I. **Data Isolation:** We verified that Row Level Security (RLS) policies are active. A patient authenticated with Token A cannot query the data of Patient B.

II. **No PII in LLM:** We ensured that while health metrics are sent to the LLM, no Personally Identifiable Information (PII) like real names or addresses are included in the prompt context to preserve privacy.

7. Lessons Learned (Critical Review of Sprint Two Experience and Future Plan)

7.1. Lessons Learned

I. **Don't over-promise on AI:** We learned that "getting the data ready" for AI takes 80% of the time. The actual LLM integration is fast, but building the pipeline to feed it clean data is a slow process.

II. **State Management is Key:** In a data-heavy health app, centralised state management (Riverpod) is not optional, it is essential to prevent UI bugs.

III. **Authentication is Complex:** Deep linking and session management on mobile devices require significantly more testing time than web-based auth.

7.2. Recommendations for Future

I. **Freeze Core Features:** No new "core" features should be added. The focus must now shift entirely to polishing and documentation.

II. **Clinician UX:** The next immediate task is to smooth out the clinician's user journey to match the quality of the patient dashboard.

7.3. Plan for Next Semester

With the foundation complete, the roadmap for the next semester is clear:

I. **Implement the LLM Automation Engine:** Now that we have reliable data ingress, we can build the background workers to trigger alerts.

II. **Advanced Analytics:** Move from simple line charts to predictive trend analysis.

III. **Final Polish:** Comprehensive error handling and UI animations.

This sprint successfully redeemed the project's trajectory by delivering a fully integrated, demonstrably working product.